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Impacts of climate change on groundwater quality: a systematic literature review of analytical models and machine learning techniques

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E-mail: md.salauddin@ucd.ie, tahmida.n.chowdhury@ucdconnect.ie, abattamo@ucc.ie, raj.nag@ucd.ie and Ivar.Zekker@ut.ee**Keywords:** contamination, groundwater quality index, GIS, MODFLOW, machine learning prediction

Abstract

The increasing impact of global climate change on hydrogeological and hydrological systems presents substantial challenges to the sustainable management of groundwater quality (GWQ). Changes in precipitation regimes, temperature fluctuations, and the frequency of extreme hydro-climatic events driven by climate change accelerate the deterioration of GWQ, thereby threatening ecosystems and human health. In response to these challenges, recent research has increasingly focused on developing and refining analytical models (AM) and machine learning (ML) techniques to understand better and predict the impacts of climate change on GWQ. This systematic literature review critically examines the current state of knowledge on applying AM and ML models in the context of GWQ assessment under climate-induced stressors. By synthesizing findings from a comprehensive review of existing studies, this paper discusses the capabilities, limitations, and future directions of hybrid ML and traditional AM in GWQ prediction, vulnerability, and threshold estimation. The review reveals that while ML approaches significantly enhance predictive accuracy and model robustness, there remain substantial challenges in their application due to the complexity of climate-induced variables and the scarcity of high-resolution data. This paper aims to provide GWQ researchers, water resource managers, and policymakers with an advanced understanding of the interactions between climate change and GWQ and the innovative AM and ML modelling approaches available to address these challenges. By highlighting the potential and limitations of current models, this review offers insights into developing more effective and adaptive management strategies for safeguarding GWQ in an era of rapid climatic change.

1. Introduction

The ecosystems often maintained by direct or indirect groundwater supplies are collectively known as groundwater-dependent ecosystems (GDEs) and are critical for sustaining biodiversity (Eröstate *et al* 2020, Chen *et al* 2023). However, climate change significantly impacts groundwater ecosystems, leading to

alterations in groundwater quality (GWQ) and availability. The overall climate-induced changes in GWQ include changes in pH, dissolved oxygen level, dissolved organic matter, salinity, microbial community, redox status, and concentrations of organic and inorganic compounds (Saccò *et al* 2022, Houghton *et al* 2023, Dao *et al* 2024). GWQ is also influenced by changes in climatic factors such as increasing

temperature, changing precipitation patterns, and sea level rise, as well as extreme weather events such as heavy precipitation, floods, droughts, and storms (Dao *et al* 2024). The potential impacts of climate change on the quality and quantity of groundwater include changes in recharge rate, groundwater storage, discharge rate, and GWQ (Lindquist *et al* 2019, Gintamo *et al* 2021, Neidhardt and Shao 2023, Dao *et al* 2024). Understanding these impacts is an important step in developing appropriate groundwater management strategies.

Groundwater is considered a high-quality freshwater resource for drinking, irrigation and industrial purposes in several parts of the world (Chen *et al* 2023) which led to the rapid depletion of this valuable natural resource. The groundwater ecosystem has deteriorated due to notable alterations in climate and land use in recent years (Wang *et al* 2021). Climate change exerts considerable pressure on water resources, mainly in groundwater, because of increased temperature and precipitation (Wang *et al* 2021). The impacts of increasing temperature on GWQ include changes in primary productivity and microbial activity, alteration of geochemical processes, harmful algae blooms, increased pesticide use, and elevated wildfire risk (Dao *et al* 2024). According to research, shallow groundwater temperatures have responded quickly to recent climate change, which sheds light on how vulnerable aquifers and ecosystems that depend on groundwater are to future climate change (Menberg *et al* 2014). Additionally, changes in rainfall patterns due to climate change substantially impact groundwater microbial communities. The effects of changing precipitation regime on GWQ include changes in soil hydraulic conductivity, variations in runoff flux, dilution effects, and increased cracking of landscapes, leading to more filtration (Gintamo *et al* 2021, Šperac and Zima 2022, Dao *et al* 2024). Apart from that, the impacts of sea level rise on GWQ include altered geochemical processes and changes in microbial communities (Saccò *et al* 2022, Houghton *et al* 2023, Dao *et al* 2024). For example, recent research by Saccò *et al* (2021) highlighted that the high rainfall triggered the taxonomic diversity of microbial communities such as copepods and amphipods. The groundwater microbial ecosystem is negatively impacted by increasing chloride concentrations, including declines in bacteria responsible for nitrate and ammonia removal (Houghton *et al* 2023), a consequence of sea level rise. Moreover, the excessive expansion of human activities poses a significant threat to the long-term viability of coastal GDEs and their services. Additionally, coastal areas need more legal management acknowledgement to protect the coastal groundwater ecosystems (Eröstate *et al* 2020). This is because rising sea levels and temperatures exacerbate already-existing climate-related hazards, such as storms, coastal and land floods, and

coastal erosion, making coastal areas more susceptible to the effects of climate change and demanding focused attention (Laino and Iglesias 2024). A heightened focus is also required to avoid extensive groundwater extraction in the coastal aquifers as extensive withdrawal exacerbates saltwater intrusion, posing risks to coastal vegetation and freshwater supplies (Felisa *et al* 2013). Hence, it is clear that climate change substantially influences the quality of groundwater, and it is crucial to analyse this impact. In this regard, modelling facilitates the most accurate assessment and prediction of quality parameters and contamination. Groundwater modelling is traditionally based on mathematical process-based physical models (PBPMs) for GWQ assessment and prediction (Mokarram *et al* 2019, Cheng *et al* 2024, Huang *et al* 2024). Particularly, assessing the effect of climate change on GWQ is a complex technical task (Gintamo *et al* 2021). Mathematical models, such as Analytical Hierarchy Process (AHP) (Chorol and Gupta 2023a, Cheng *et al* 2024), Hydrological modelling (Arnold *et al* 1998, Nassery and Salami 2016, Dash *et al* 2021, Ngo *et al* 2024) and geographic information system (GIS)-based drastic model (Abu-Bakr 2020, Balaji *et al* 2021, Torkashvand *et al* 2021, Chakraborty *et al* 2022) have been widely used in identifying vulnerable areas or hotspots (Nassery and Salami 2016, la Cecilia *et al* 2020). However, traditional approaches assume the existence of a linear relationship between the dependent and independent variables; hence, they cannot adequately capture the complex relationships among attributes and have the disadvantage of being computationally demanding (Huang *et al* 2024, Meray *et al* 2024). In recent years, machine learning (ML) models have been suggested as a viable alternative for extracting valuable information from data. This approach is being increasingly implemented due to the reliability and efficiency of ML, where classical PBPMs were previously employed (Haggerty *et al* 2023, Huang *et al* 2024).

With the advancement of ML models, researchers in recent years have employed data-driven techniques for predicting GWQ. For example, ML algorithms have been used in mapping GDEs (Martínez-Santos *et al* 2021), groundwater contamination, mostly nitrate (Judeh *et al* 2022) and GWQ parameters (Sahour *et al* 2023). Martínez-Santos *et al* (2021) used the supervised classification algorithm for mapping GDEs. They predicted the presence of GDEs with the accuracy of >96% test and AUC scores. Another study in the USA by Rohde *et al* (2021) adopted the Random Forest (RF) algorithm with satellite-based remote sensing to predict groundwater levels (GWLs) under GDEs. The result revealed that GDEs are experiencing a long-term decline in GWLs. Additionally, Judeh *et al* (2022) used ML algorithms, GIS, and statistical approaches for GWQ management and nitrate contamination. Numerous other researchers analysed and predicted the GWQ parameter using popular

ML algorithms. The most widely used ML models in GWQ predictions are Decision Trees (DTs) (García-del-Toro *et al* 2022, Venkatesh *et al* 2024), Extreme Gradient Boosting DTs (XGBoost) (Alshehri and Rahman 2023, El-Rawy *et al* 2024, Solangi *et al* 2024, Sunita and Ghosh 2024), and RF (Singha *et al* 2021, Sahour *et al* 2023, El-Rawy *et al* 2024, Zheng *et al* 2024). Moreover, kernel-based methods such as support vector machine (SVM) (Shiri *et al* 2021, Lap *et al* 2023, Sahour *et al* 2023) and artificial neural networks (ANNs) (Shiri *et al* 2021, Elmeddahi and Ragab 2022, Kouadri *et al* 2022) are the other popular ML models in groundwater research.

Additionally, it is becoming increasingly popular to use the hybrid model in GWL and quality predictions. Hybrid models combine the strengths of different modelling techniques, such as ML models and traditional analytical models, to enhance the accuracy, performance and reliability of GWQ predictions. These models are particularly useful in scenarios where complex relationships exist between input variables and GWQ and where data-driven models alone may not provide satisfactory results. For instance, Roshni *et al* (2019) demonstrated the use of a hybrid model specifically combining feedforward ANNs with a wavelet ANN model, and further enhanced by the Gamma and M-tests (GT) approach. This hybrid model is used to forecast spatio-temporal groundwater fluctuations in complex aquifer systems, offering a robust predictive tool for hydrological processes. Another study (Jibrin *et al* 2024) integrated Gaussian process regression (GPR), adaptive neuro-fuzzy inference system (ANFIS), and DTs, which provides an innovative approach for predicting the Water Quality Index (WQI) and GWQ Index (GWQI). This model is particularly useful in regions with limited or no monitoring data, such as arid and semi-arid areas like Saudi Arabia, where traditional monitoring systems may be sparse.

On the other hand, analytical models have also been employed in a significant amount of research (Jha *et al* 2020, Singh *et al* 2021, Li *et al* 2023, Srivastava *et al* 2024). The conventional weighted index method, multivariate statistical methods, fuzzy logic and fuzzy AHP methods have been successfully used in assessing groundwater vulnerability and quality. However, existing GWQ assessment methods face challenges due to the random and nonlinear nature of water quality (WQ) indicators. While fuzzy comprehensive assessment helps handle ambiguity, its results are often subjective due to expert-assigned weights. The WQI method provides both quantitative and qualitative evaluation but suffers from uncertain indicator weights. Multivariate statistical analysis reduces dimensionality but is limited in classification accuracy (Li *et al* 2023). In such scenarios, ML models tend to outperform analytical models in terms of

prediction accuracy when the system is complex and data-rich. They excel at capturing complex patterns and interactions, making them well-suited for predicting GWQ in diverse environments.

Thorough reviews comparing analytical models and ML modelling for GWQ analyses are limited. Although Afrifa *et al* (2022) reviewed the application of mathematical and ML models to predict changes in GWLs, the study did not address GWQ.

Apart from that, a few literature review papers have been published on the applications of ML for modelling GWQ. For instance, (Haggerty *et al* 2023) reviewed the application of supervised, semi-supervised, unsupervised, and ensemble ML models to predict GWQ parameters. The topic of the review is broad, and the discussion of ML applications on the impacts of climate change on GWQ has been less thoroughly covered. Additionally, more recently, Bamal *et al* (2024) demonstrated the effectiveness of several ML/AI models in predicting groundwater parameters, including levels, outflow, storage, and quality, under diverse climatic conditions such as temperature and precipitation that affect these variables. However, the study lacks the necessary focus and comprehensiveness about the use of analytical and ML methodologies in GWQ modelling and prediction. Therefore, it appears that both AM and ML have been applied to predict groundwater vulnerability and quality in numerous studies (Haggerty *et al* 2023, Ibrahim *et al* 2023, Sahour *et al* 2023, Huang *et al* 2024); nevertheless, there is a lack of systematic literature reviews on both AM and ML techniques for GWQ modelling considering the impacts of climate change.

This study conducted a systematic literature review of AM and ML in GWQ modelling in terms of climate change on groundwater contamination. In comparison with existing literature reviews (Haggerty *et al* 2023, Bamal *et al* 2024, Dao *et al* 2024), this review makes four main contributions. Firstly, the study presents an overview of the potential risks of GWQ posed by global and regional climate change. Secondly, it provides the most up-to-date and relevant information about the applications of analytical and ML techniques to GWQ modelling, thereby highlighting their relevance and potential in practical application. Thirdly, it reviews the comparative application of analytical and ML-based modelling approaches employed in GWQ analysis and, finally, discusses the implementation of the hybrid approach in different studies.

This research aimed to address the following specific questions to fulfil the objectives.

- (i) What are the potential hazards to the groundwater ecosystems caused by groundwater contamination and climate change?

- (ii) How can analytical models assist in identifying vulnerable areas or hotspots where climate change and contamination are likely to have the greatest impact on GWQ and ecosystems?
- (iii) How effectively do ML algorithms predict GWQ parameters?
- (iv) How can hybrid modelling approaches, integrating analytical and ML methods, elucidate the complex interactions between climate change and GWQ?

By answering these questions, this review provides the most up-to-date and relevant information about applying analytical and ML modelling approaches to GWQ, highlighting its relevance and potential in practical application.

2. Methodology

This study followed a comprehensive systematic literature review approach to provide a deep understanding of climate change impacts on groundwater ecosystems. A well-designed approach with every existing research detail in a particular field is necessary to identify future research potentialities and further improvements with the current state of knowledge (Uc-Castillo *et al* 2023). A systematic review aided the authors in preparing transparent and complete research. It revealed the answers to why the review was done, what they did, and what they found. The updated 2020 Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) statement is employed in this study for the detailed review process (Page *et al* 2021). Many authors have successfully employed the PRISMA statement for systematic review in previous review studies related to groundwater management, ML and other areas (Braga *et al* 2020, Kondash *et al* 2021, De Paula and Marques 2022, Ghobadi Fatemeh 2023, Uc-Castillo *et al* 2023). This detailed review method followed four steps: (1) identification, (2) screening, (3) eligibility, and (4) inclusion (Alfadil *et al* 2022, Uc-Castillo *et al* 2023). The PRISMA flow diagram (Page *et al* 2021) summarizes the procedure employed in this study (figure 1).

2.1. Identification

Using the PRISMA conceptual framework this systematic review performs a comprehensive search strategy to identify the relevant articles from prominent research databases, including Scopus, Web of Science and Google Scholar. The search is limited to publications published only in the English language after 2000, aiming to explore the impacts of climate change and contaminations on GWQ and ecosystems with the application of ML. The search strategy followed a designed search string containing specific

terms, including ‘contamination’, ‘groundwater ecosystems’, ‘climate change’, ‘ML’, and ‘analytical models’. In total, the search yielded 441 articles across all selected databases, demonstrating the broad scope of literature available on this topic. The details of the search string can be found in table 1.

2.2. Screening

Following the search of relevant research publications, information, for example, article title, keywords, authors, publication year, digital object identifier and other details, were extracted and prepared in a Microsoft Excel spreadsheet. There are duplicates for using two databases and in the same databases with improvised search strings. The duplicates were removed using the Excel duplicates command which resulted in 399 articles (figure 1) to be used for next step screening. In this study, only primary research articles, review papers and conference papers were considered for the review process. These articles were then assessed for inclusion and exclusion criteria. After evaluating the title and abstract 144 papers were removed, and they were evaluated individually for the eligibility of the remaining articles. In cases of uncertainty, article titles were submitted to external reviewers who were proficient in the relevant discipline for assessment. Consensus among the evaluators or, if necessary, the suggestions from external experts served as the determining factor in the final decision to include or exclude an article. The articles were chosen with the intention of broadening the scope by considering the interdisciplinary areas, with the exception of any direct irrelevancies that were identified in the included research areas.

2.3. Establishment of eligibility criteria

(a) Inclusion criteria:

- (i) Articles that focus on Climate change, GWQ and ecosystems and ML.
- (ii) Research areas including water resources, environmental sciences, ecology, geology and others (figure 2(c)).
- (iii) Scholarly publications written in English from 2000 to 2024 to encompass current progress and developments.
- (iv) Articles and conference proceedings with original research findings and previous relevant review papers.

(b) Exclusion criteria:

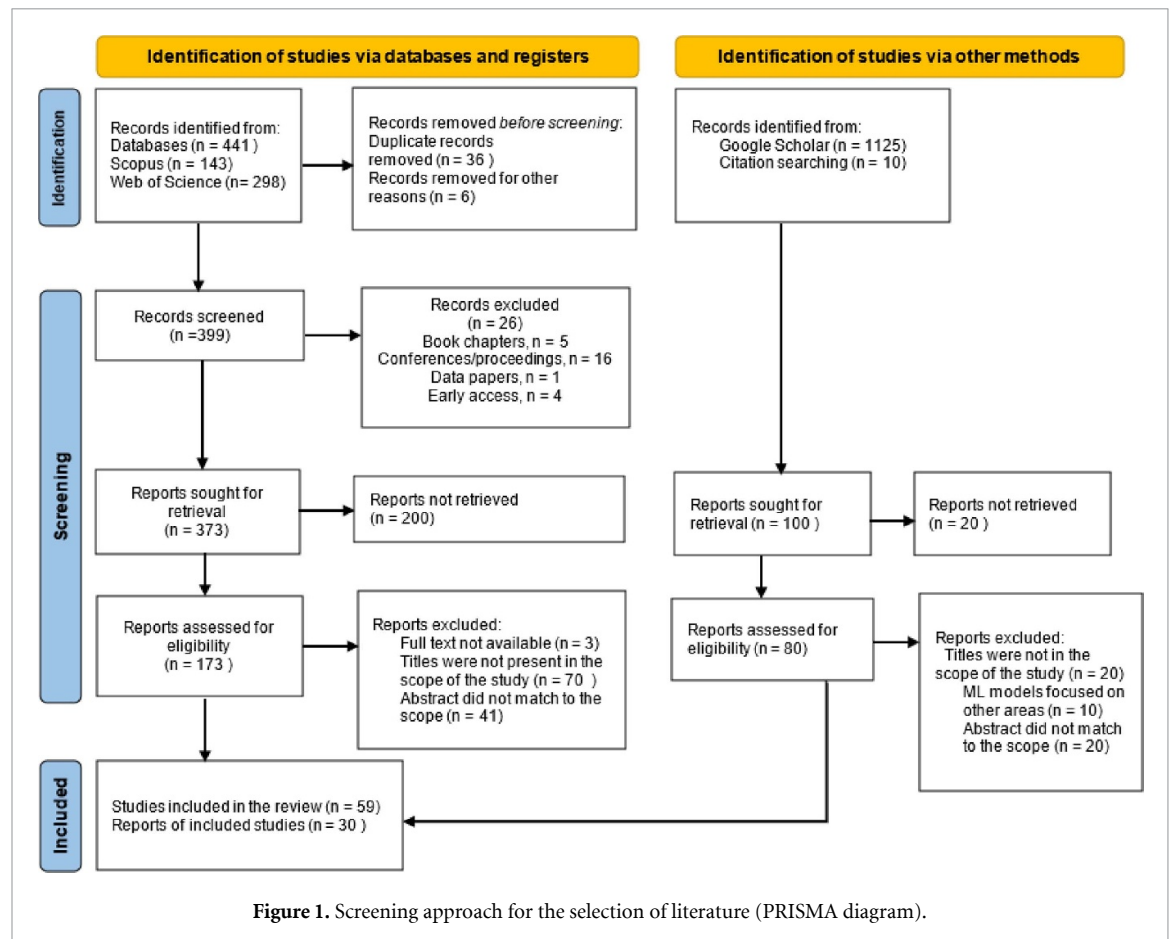
- (i) Research areas that are not related to this study such as social science, material science and others.
- (ii) Book chapters, conference abstracts, data papers and early access

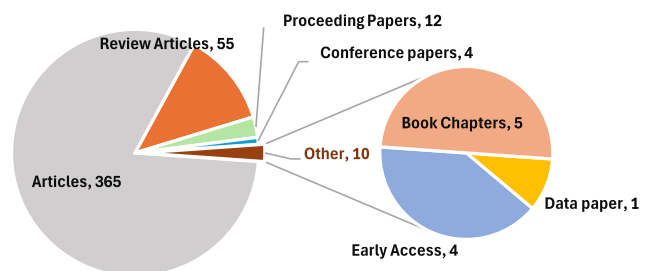
Table 1. Search string for Scopus and Web of Science databases.**Scopus search string**

- TITLE-ABS-KEY ({Climate change} AND {groundwater ecosystems}) AND (EXCLUDE (SUBJAREA, 'SOCI') OR EXCLUDE (SUBJAREA, 'MEDI'))
- TITLE-ABS-KEY ({groundwater} AND {ecosystems} AND {climate change} AND {machine learning}) AND (EXCLUDE (SUBJAREA, 'SOCI') OR EXCLUDE (SUBJAREA, 'BUSI') OR EXCLUDE (SUBJAREA, 'MEDI') OR EXCLUDE (SUBJAREA, 'ENER'))
- TITLE-ABS-KEY ({groundwater} AND {ecosystems} AND {analytical model}) AND (EXCLUDE (SUBJAREA, 'DECI') OR EXCLUDE (SUBJAREA, 'ECON') OR EXCLUDE (SUBJAREA, 'MATH')) AND (EXCLUDE (LANGUAGE, 'Chinese'))
- TITLE-ABS-KEY ({groundwater} AND {ecosystems} AND {machine learning}) AND (EXCLUDE (SUBJAREA, 'DECI') OR EXCLUDE (SUBJAREA, 'MATH') OR EXCLUDE (SUBJAREA, 'BUSI') OR EXCLUDE (SUBJAREA, 'MEDI') OR EXCLUDE (SUBJAREA, 'PHYS') OR EXCLUDE (SUBJAREA, 'SOCI') OR EXCLUDE (SUBJAREA, 'PHAR') OR EXCLUDE (SUBJAREA, 'ENER'))
- TITLE-ABS-KEY ({contamination} AND {groundwater ecosystems}) AND (EXCLUDE (SUBJAREA, 'MEDI') OR EXCLUDE (SUBJAREA, 'DECI') OR EXCLUDE (SUBJAREA, 'SOCI') OR EXCLUDE (SUBJAREA, 'PHAR'))
- TITLE-ABS-KEY ({groundwater} AND {ecosystems} AND {climate change} AND {analytical model})
- TITLE-ABS-KEY ({temperature} AND {groundwater ecosystems}) AND (EXCLUDE (SUBJAREA, 'IMMU') OR EXCLUDE (SUBJAREA, 'SOCI') OR EXCLUDE (SUBJAREA, 'MEDI')) AND (EXCLUDE (LANGUAGE, 'German'))

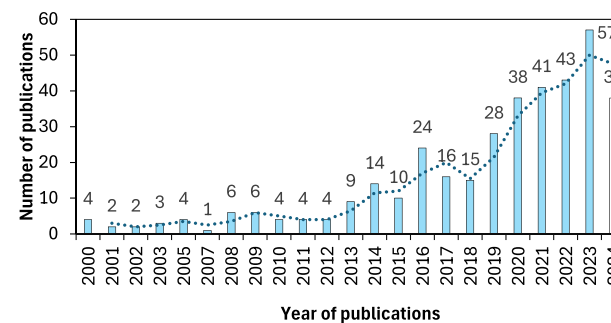
Web of Science search string

- ((TS = climate change (Topic) AND contamination (Topic) AND groundwater ecosystems (Topic)
- ((TS = climate change (Topic) AND groundwater quality (Topic) AND Machine learning (Topic)
- ((TS = climate change (Topic) AND groundwater quality (Topic) AND analytical modelling (Topic)
- ((TS = climate change (Topic) AND groundwater quality (Topic) AND hybrid modelling (Topic)
- ((TS = Uncertainties (Topic) AND probabilistic modelling (Topic) AND stochastic modelling (Topic) AND groundwater vulnerability (Topic)
- ((TS = climate change (Topic) AND spatial and temporal changes (Topic) AND groundwater resources (Topic) AND machine learning (Topic)
- ((TS = climatic temperature (Topic) AND groundwater ecosystems (Topic)

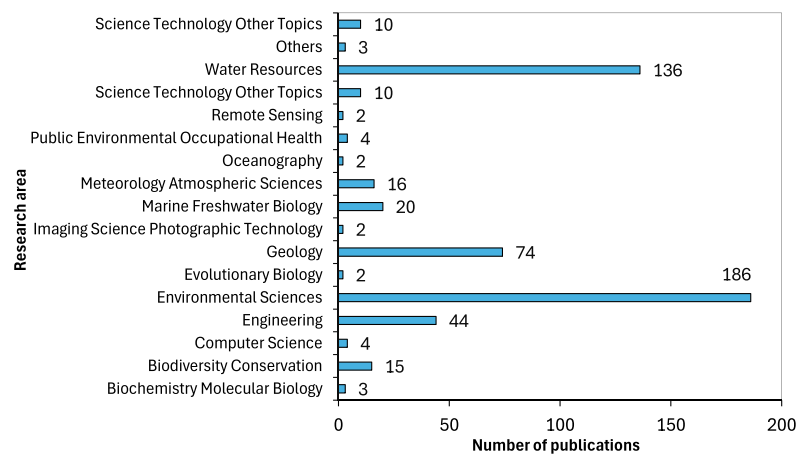




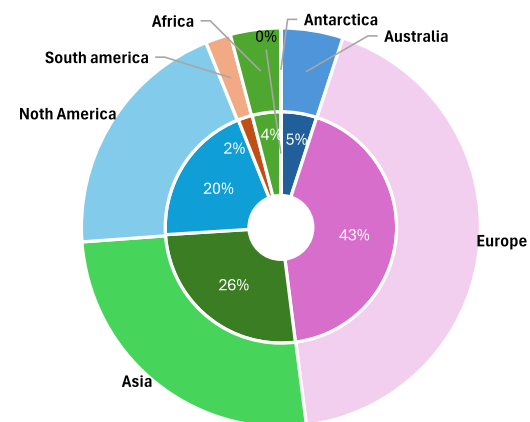
a) Number of Publications according to document type



b) Number of Publications in different years



c) Number of Publications by Subject Areas



d) Number of Publications on different continents

Figure 2. A summary of systematic review and current diversity of literature in a nutshell.

- (iii) Articles published before 2000 to maintain focus on recent developments.
- (iv) Publications that are not available in English or lack sufficient information.

3. Results and discussion of literature search

The subsequent section discusses publication trends, which encompass the number of different types of research articles, publications according to years, the number of journals in different research areas, and geographic location.

3.1. Search results

Primary identification (figure 2(a)) results in a total of 446 articles containing research articles, review articles, conference papers, book chapters and data papers. The highest number of publications were research articles comprising 365 numbers, while the lowest report was 1 data paper. The second highest number of publications were review articles (55). Apart from that, the total number of proceeding and conference papers was 12 and 4, respectively. Among the other categories, the search string identified 5 book chapters and 4 early access papers. However, this study only covers the research articles and review articles in the domain of groundwater and ML.

3.1.1. Publication trends by year

The number of yearly publications retrieved from the search results is depicted in figure 2(b). Research articles on the ML approach for the impact of climate change and contamination of groundwater have increased throughout the period from 2000 to 2024. It is noticed that a maximum number of articles were published in 2023 which is 57. Figure 2(b) shows the number of articles published between 2000 and 2012 was sparse, where, on average, five articles were published in that period. After that period, the publication trends fluctuated till 2018 before beginning to increase and reaching a peak in 2023. It is observed that ML techniques have become popular in recent years for predicting GWLs and water resource management, which was also reviewed by a few authors (Ghobadi Fatemeh 2023, Uc-Castillo *et al* 2023) recently.

3.1.2. Research article by subject area

Figure 2(c) further illustrates the field-wise contribution of the publications. The findings indicated that most of the journals (186) belong to environmental science. With a total of 136 articles, water resources constitute the second most prolific contributory field. Geology and engineering contributed to a total of 74 and 44 publications, respectively, while the contribution of other fields was ≤ 20 .

3.1.3. Geographic location

The articles from different countries, including the continents of Europe, Asia, America, Australia, Africa, and Antarctica, are illustrated in figure 2(d) and appendix (table A1 and figure A1). The search showed that Europe (43%), Asia (26%) and North America (20%) are the three continents with the highest number of publications (figure 2(d)). While the lowest number of publications are published in Africa and South America which are 4 and 2% respectively.

4. Groundwater and climate change

A growing number of elements, both climatic and non-climatic, have been posing threats to GWQ since the dawn of the modern age. Air and ocean temperatures have risen in recent decades, resulting in scorching days, warm nights and heat waves. In the same way, regular amounts of rain and snow have gone up in high latitudes while declining in subtropical, intermediate, and lower latitudes (Aslam *et al* 2018). Therefore, it is no longer possible to ignore the effects of climate change-induced warming on our ecosystems; nevertheless, our understanding of the effects on groundwater ecosystems and GWQ is limited. Because of their extreme site-specificity and regional heterogeneity, warming's effects on groundwater are difficult to quantify and anticipate (Neidhardt and Shao 2023).

Groundwater is a nearly global source of fresh water that is usually of very high quality. One-third of all freshwater withdrawals worldwide comes from groundwater, which provides an estimated 36%, 42%, and 27% of the water needed for residential, commercial, and agricultural uses, respectively (Taylor *et al* 2013). On a global scale, groundwater systems are being fundamentally impacted by human activities, which may manifest directly as pumping and overuse or indirectly as climate change-induced alterations in terrestrial hydrology (Stigter *et al* 2023).

4.1. Potential hazards caused by groundwater contamination and climate change to groundwater ecosystems

Subsurface ecosystems that store the world's largest reservoirs of liquid freshwater are highly vulnerable because of climate change. Extreme hydrological events, such as long-term droughts, massive rainfall, and rising temperatures, change the accessibility and quality of groundwater. Moreover, increasing water demand exacerbates the pressure on already fragile groundwater systems. These alterations have an impact on the carbon and nitrogen cycles and other ecological processes by disrupting the nutrient-poor aquifers that support diverse microbial and faunal communities (Retter *et al* 2021). On the other hand, a significant amount of the formerly pure aquifers are now contaminated due to a variety of pollutants

that come from various sources throughout the world. Chemical and hydrogeological characterization of aquifers and contaminants has been ongoing since the 1950s, intending to understand their potential impacts on human health and, more lately, on GDEs, for example, springs, marshes, and swamps (Morrissy *et al* 2022). Over the past few years, numerous aquifers have seen a growing level of contamination due to various human activities. One of the most widespread is nitrogen contamination for agricultural activity (McAleer *et al* 2022, Morrissy *et al* 2022). Agricultural application of nitrogen is a potential source of groundwater nitrate contamination via manures, fertilizers and stock excretion. This inorganic nitrogen in the form of NO_3^- reaches groundwater through leaching when it coincides with a high recharge system (McAleer *et al* 2022). Regarding how climate change affects groundwater, numerous research (Green *et al* 2011, Taylor *et al* 2013, Ducci and Polemio 2018) have demonstrated that there are significant consequences for both WQ and quantity.

4.1.1. Precipitation

Climate change has a profound impact on precipitation patterns, which can cause severe drought and lower the GWL during extreme conditions. Variations in precipitation have an impact on GWQ as well because they load nutrient-rich urban runoff into aquifers (Gintamo *et al* 2021). GWLs are affected by precipitation changes due to extreme weather events, resulting in variations of levels of up to 4–5 m. However, the groundwater of deep alluvial aquifers remains qualitatively and quantitatively stable in the variation of precipitation during severe climate conditions, thereby ensuring the stability of the public water (Šperac and Zima 2022). According to the analysis of climate extremes by Šperac and Zima (2022), the years 2000, 2003, and 2011 were characterized by extreme weather conditions, with droughts lasting nearly the entire year of 2000. Conversely, simulation analyses (Gintamo *et al* 2021) forecast that yearly precipitation will rise by 2042 and subsequently decline until 2060.

4.1.2. Sea level rise

Undoubtedly, the ongoing rise in global temperatures will inevitably increase the average sea level worldwide. The coastal groundwater aquifers have significantly degraded in both quantity and quality due to alterations in recharge and rising sea levels caused by climate change. Sea-level rise (SLR) is anticipated to increase coastal erosion and inundation, even without the potential impacts of the changing climate on storm surges and waves. By 2100, even under 1.5 °C of warming, and frequently long before the end of the century, almost 50% of the 7000+ places taken into will encounter the current 100 year extreme sea-level event annually. The tropics exhibit greater sensitivity compared to the Northern high latitudes,

as certain sites do not experience a change in frequency even under the most extreme global warming scenarios (Tebaldi *et al* 2021).

4.1.3. Saltwater intrusion

As a result of both natural factors and the effects of climate change, groundwater salinization is becoming an increasingly important environmental concern in dry and semi-arid regions (Castaño-Sánchez *et al* 2020). Saltwater intrusion is a potential threat to groundwater aquifers in coastal regions (Houghton *et al* 2023). A climate scenario simulation in Kenya (Okello *et al* 2015) indicated significant susceptibility to saltwater intrusion, and the aquifer is projected to be inadequate in meeting the water demands of the growing population. In the event of a rise in sea level, the boundary between saltwater and freshwater will shift towards the land. It may influence groundwater wells previously unaffected by seawater intrusion during periods of low rainfall. An excessive amount of water pumped from these wells may also cause seawater to enter the bore (Houghton *et al* 2023).

4.1.4. Coastal flooding

The intricate interplay between natural and human-caused variables is what causes the phenomena known as global climate change. As a result of human actions like greenhouse gas emissions and increasing atmospheric gas concentrations, climate patterns become unbalanced. This scenario impacts the frequency and intensity of climate events, where floods being one of them (Gül and Kayaalp 2024). According to the Intergovernmental Panel on Climate Change (IPCC) (2021), climate change forecasts indicate that there will be a persistent increase in global sea levels throughout the end of the twenty-first century and beyond. Coastal areas are recognized as regions that face significant threats, as indicated by the IPCC in 2022. Coastal areas close to sea level are especially susceptible to risks from coastal hazards. Both sea level rise and severe rainfall events will progressively exacerbate flooding until the end of the century (Bormann *et al* 2024). Xu *et al* (2023) conducted a study on the coastal island of Hainan, China, revealing that climate change will result in increased rainfall and water levels and heightened flood levels, depth, and duration. Researchers have also performed groundwater inundation mapping to provide decision-makers with data regarding regions with an increased likelihood of flooding. In the short term, uncertainty in sea-level rise projections contributes relatively little to the uncertainty of groundwater flooding probabilities, with history matching effectively reducing uncertainty in near-term predictions. However, as predictions extend further into the future, SLR uncertainty becomes the dominant factor influencing groundwater flooding predictions (Chambers *et al* 2023). Therefore, groundwater modelling under changing climate scenarios

is indispensable for understanding the interplay between rising sea levels, altered hydrological cycles, and GWQ. By leveraging probabilistic frameworks and history matching, models not only predict risks but also reduce uncertainty, providing actionable insights.

4.1.5. Temperature changes

The effects of climate change-induced warming on ecosystems are increasingly significant. However, our comprehension of its implications for groundwater ecosystems and GWQ remains profoundly inadequate (Neidhardt and Shao 2023). Groundwater temperatures are intricately associated with land surface temperatures, and shallow groundwater temperatures have quickly reacted to climate change (Menberg *et al* 2014). A notable temperature rise of 1.9 °C–2.3 °C will impact GWQ by enhancing evaporation and elevated salt concentrations (Gintamo *et al* 2021). An analytical solution by Kurylyk and Macquarrie (2014) revealed that groundwater in the subsurface can be heated by recharge, while upward discharge can further increase warming in the deeper subsurface. The air temperature (AT) can also impact the temperature of groundwater. An analysis by (Egidio *et al* 2022) shows a clear correlation between the mean minimum AT and the mean minimum GWT in an unconfined aquifer in the Campanian Plain, Southern Italy. Climate change is also raising winter temperatures globally, hence modifying snow-melt hydrology in the region of higher altitudes. (Ford *et al* 2021). It is anticipated that average annual and monthly temperatures will rise substantially by the end of the twenty-first century. The rise in winter temperatures will diminish snow precipitation and promote the melting of the snow cover in the uppermost sub-basins (Pisani *et al* 2019).

4.1.6. Effects on GW-dependent organisms

It is anticipated that climate change will exacerbate the need for underground water due to an increased occurrence of droughts and anthropogenic influences on groundwater and its dependent ecosystems (Di Cicco *et al* 2023, Stigter *et al* 2023). Specialized organisms that can live and flourish in habitats that are usually energy- and oxygen-limited but thermally stable are found in groundwater ecosystems (Di Cicco *et al* 2023). Di Cicco *et al* (2023) focused on *Niphargus longicaudatus*, a crustacean amphipod, to determine its thermal niche breadth using the chronic method. Results show *N. longicaudatus* thrives in temperatures ranging from 7 °C to 9 °C, similar to its Italian distribution range. Surprisingly, its preferred temperature range differs from mean annual temperatures at collection sites, challenging assumptions about species' adaptability to narrow temperature ranges. They suggested that understanding these species' fundamental thermal niches can guide conservation and management

efforts to safeguard groundwater ecosystems and their inhabitants. Another study conducted by Castaño-Sánchez *et al* (2020) tested the temperature and salinity tolerance of groundwater-adapted invertebrates, focusing on representative crustaceans from Australia. Their result revealed that copepods and syncarids experienced mortality at specific salt concentrations and temperature increases. Copepods were more sensitive than syncarids, with a predicted risk of population loss under future global warming scenarios. Moreover, groundwater microbial communities experience varied effects from increasing chloride concentrations. This increase in chloride concentrations leads to a decline in bacteria crucial for nitrate and ammonia removal, disrupting nutrient cycles revealed by Houghton *et al* (2023). These ecosystems are particularly vulnerable to threats such as sea level rise and groundwater extraction for agriculture. Therefore, protecting these ecosystems is essential to maintaining their ecological balance, especially against anthropogenic pressures. Apart from that, in a recent review work (Becher *et al* 2022), it is mentioned that groundwater fauna (stygo fauna), which play a vital role in ecosystem services within oxygenated aquifers, are highly sensitive to anthropogenic impacts, particularly in urban areas. These organisms, with their slow metabolisms and long life cycles, face significant threats from urban pollution, temperature increases and salinity changes. These findings emphasize the dangers posed by rising salinity and temperatures to the ecological health of groundwater ecosystems. The key climate factors and their impacts on GWQ are summarized in table 2.

4.1.7. WQ impacts on ecosystems in a changing climate
Groundwater ecosystems, which are inhabited by a variety of microbial communities, are exceedingly susceptible to alterations in WQ. The challenges that these ecosystems face, intensified by human activities and climate change, are summarized in table 3.

4.1.7.1. Nitrogen contamination

The increasing use of nitrogen is threatening GWQ and its dependent ecosystems. Over the last century, reactive nitrogen (N) fluxes have increased tenfold due to population growth, dietary changes, and widespread use of N fertilizers (Byrnes *et al* 2020). Agricultural activities are a significant source of groundwater nitrate contamination in numerous regions of the globe and are frequently referenced in numerous studies (Byrnes *et al* 2020, Sarkar *et al* 2022, Liu *et al* 2024, Zhang *et al* 2024). Runoff from agricultural landscapes introduces excess nitrogen into groundwater, threatening WQ and altering ecosystem dynamics. Although the N inputs are managed, however, the accumulation sources, such as soil and groundwater reservoirs, continue to impact WQ. On the other hand, climate change exacerbates nitrogen-related issues through increased

Table 2. Key impacts of climate change on groundwater quality.

Climate factor	Impact on groundwater quality	Affected parameters	Regions affected	References
Increased temperature	Increases evaporation, affects recharge rates	Salinity, Nitrate, pH	Arid & semi-arid regions	(Neidhardt and Shao 2023)
Changing precipitation	Alter recharge and dilution rates	Contaminants, Pathogen levels	Mediterranean, Tropical regions	(Gintamo <i>et al</i> 2021)
Sea level rise	Causes saltwater intrusion	Salinity, total dissolved solids	Coastal aquifers	(Tebaldi <i>et al</i> 2021)
Extreme weather events	Increases flooding, contaminant runoff	Organic pollutants, heavy metals	Flood-prone areas	(Gül 2024; Xu <i>et al</i> 2023)
Decreased snowpack	Reduces spring recharge, affecting the water table	Nutrients, pesticides	High-altitude regions	(Ford <i>et al</i> 2021; Lindquist <i>et al</i> 2019; Pisani <i>et al</i> 2019)

Table 3. Influence of climate change on key groundwater quality parameters (observed trends from studies).

Parameter	Observed impact	Associated climate factor	Study region	References
Nitrate	Increase due to reduced dilution	Decreased precipitation	Southern Europe	(McAleer <i>et al</i> 2022)
pH	Lowered due to increased acidification	Rising temperature	Southeast Asia	(Li <i>et al</i> 2023; Sunita and Ghosh 2024)
Salinity	Increase in saltwater intrusion	Sea level rise, coastal flooding	Coastal aquifers in the US	(Plane <i>et al</i> 2019)
Heavy metals (e.g. fluoride, arsenic)	Increase due to changes in oxidation state	Changing groundwater levels	South Asia	(Chorol and Gupta 2023a; Tan <i>et al</i> 2020)
Pathogen contamination	Increase due to flooding	Increased extreme weather events	Tropical regions	(Dzodzomenyo <i>et al</i> 2022)

precipitation variability, enhancing runoff and leaching. As a result, GDEs face long-term changes due to sustained nitrogen infiltration and reduced WQ (Byrnes *et al* 2020). Even the WQ in the European Alps is also negatively affected by the application of fertilizer in the grassland for increased productivity. Inefficient nitrogen uptake by vegetation, especially in high-altitude areas with short growing seasons, results in substantial nitrate leaching into groundwater. Additionally, elevated nitrate levels threaten aquatic and groundwater ecosystems, disrupting microbial communities and increasing the risk of eutrophication. In changing climate scenarios, lower temperatures and shorter growing seasons at higher altitudes reduce nitrogen uptake efficiency, leading to greater nitrate losses (Botter *et al* 2021).

4.1.7.2. Heavy metals contamination

Heavy metal pollution is another concern for GWQ because of its high toxicity, where the major source of pollution is mainly anthropogenic. Among the heavy metals, fluoride (Araya *et al* 2022, Kumar *et al* 2022, Chorol and Gupta 2023a), iron (Huynh *et al* 2022, Guo *et al* 2023), arsenic (Tan *et al* 2020, Erickson *et al* 2021), and manganese (Huynh *et al* 2022) are commonly analysed in different studies as a contamination source of groundwater. This research has focused on estimating risk due to concentration in exceeding

amounts using ML and geospatial analyses. In terms of health risks, prolonged exposure to high fluoride levels leads to dental fluorosis, a condition marked by tooth discolouration and damage, which is especially prevalent in children. Additionally, high fluoride concentrations disrupt the natural balance of groundwater ecosystems by affecting microbial communities essential for nutrient cycling and water purification (Araya *et al* 2022). A study by Kumar *et al* (2022) in India revealed that 38.50% of the groundwater in the study area is at risk of fluoride contamination. Rising global temperatures and shifting weather patterns are leading to more frequent and prolonged droughts in many regions. As a result, the amount of rainfall in certain areas is decreasing or becoming more erratic. Reduced rainfall means less dilution of contaminants such as fluoride in the groundwater, causing concentrations to rise (Thapa *et al* 2017). Moreover, Heavy metals such as Cr, Cd, and Fe can accumulate in groundwater, disrupting microbial communities essential for nutrient cycling and water purification. This leads to long-term degradation of ecosystem functions (Chorol and Gupta 2023a). In several areas of the world, geogenic arsenic contamination is also impacting GWQ. Exposure to arsenic-contaminated water is linked to severe health issues, including skin lesions, cancer, cardiovascular disease, and developmental effects in children. Additionally,

interdependencies between arsenic, phosphorus, and iron reflect complex geochemical dynamics, which are influenced by sediment deposition history and aquifer characteristics (Tan *et al* 2020). Furthermore, elevated arsenic levels disrupt microbial processes and aquatic ecosystems by altering nutrient availability and toxicity levels, potentially impacting biodiversity in groundwater-dependent systems (Huynh *et al* 2022).

5. Analytical models assist in identifying vulnerable areas or hotspots

To manage and protect water resources sustainably from quality deterioration, locating the vulnerable areas where groundwater is at risk due to contamination and climate change is important (Saranya and Saravanan 2023). In such scenarios, analytical or mathematical models are highly efficient in determining the most vulnerable areas or regions where these hazards are anticipated to exert the most substantial impact. Researchers adopted different analytical models to assess GWQ and identify the vulnerable zones concerning groundwater contamination. These models are frequently employed in a variety of groundwater studies, including the analysis of groundwater risk and contamination (Chorol and Gupta 2023a), the prediction of GWQ (Mokarram *et al* 2019), and groundwater susceptibility under different climate scenarios (Mundetia *et al* 2024). The most adopted analytical method is the multicriteria decision analysis-based AHP (Mokarram *et al* 2019, Kumar *et al* 2022, Li *et al* 2023), which is used by numerous researchers individually and integrated with other methods, including GIS (Singh *et al* 2021, Ozegin *et al* 2024), DRASTIC (Ozegin *et al* 2024), entropy weight method (EWM) and grey rational analysis (GRA) (Li *et al* 2023). In addition, hydrological models such as Soil and Water Assessment Tool (SWAT) and MODFLOW, although not exactly analytical models, were also utilized to simulate the fate and transport of nitrate contamination (Zhang *et al* 2024) in the watershed, as well as to assess groundwater drought (Bera *et al* 2024).

5.1. Weighted arithmetic mean WQI model

In general, the WQI is an advanced method that consolidates complex information to represent the overall impact of various elements on WQ, making it effective for understanding and summarizing large volumes of WQ data (El-Rawy *et al* 2024). To calculate the GWQI, generally, drinking water legislations such as the Water Framework Directive (WFD) (Directive 2000/60/EC), Groundwater Directive (Directive 2006/118/EC), World Health Organization Drinking Water Standard (WHO 2011), the American Public Health Association (APHA) Standards (APHA 2005), The Environment Conservation Rules (ECR 2023), and the Bureau of Indian Standards (BIS 2012) are

followed in several studies (Sajib *et al* 2023, Sahour *et al* 2023, Nourani *et al* 2024). Various WQI models have been employed to calculate the WQI in numerous research. The most prevalent methods are the weighted arithmetic mean WQI model (WAM-WQI) (Berhanu *et al* 2024, Krishnamoorthy and Lakshmanan 2024) and the entropy WQI model (EWQI) (Thanh *et al* 2024). The weighted arithmetic WQI (WAWQI) is the most reliable method because it relies on a single fundamental equation for a large number of quality parameters and was adopted in several studies (Elmeddahi and Ragab 2022, Raheja *et al* 2022, Apogba *et al* 2024, Berhanu *et al* 2024, Venkatesh *et al* 2024). To forecast the GWQI of a region in North Algeria, Elmeddahi and Ragab (2022) employed the weighted average approach. Additionally, Krishnamoorthy and Lakshmanan (2024) identified the water classification and WQI using this widely recognized method. The WQI model in most of the study has been developed based on several similar steps. The initial step involves the selection of WQ indicators, followed by the weighting of these indicators, then the formulation of the sub-index function, and finally, the classification of WQ. The WAWQI is:

$$\text{WAWQI} = \sum_{i=1}^n W_i Q_i \quad (1)$$

where, $W_i = \frac{w_i}{\sum_{i=1}^n w_i}$ is the relative weight of each parameter, and $Q_i = \frac{c_i}{s_i}$ is the quality rate scale obtained by dividing each parameter's concentration by the corresponding standard, w_i is the weight of each characteristic, representing its relative importance for drinking WQ on a scale from one to five, n is the number of parameters, c_i and s_i are respectively the concentration of parameter i , and the standard value of parameter i . WQ classification is an important step for assessing WQ parameters. The parameters need to be ranked from excellent to very poor. Since each WQ parameter contributes differently, different weights are assigned to the individual parameters (Sahour *et al* 2023). Generally, a weight (w_i) from 1 to 5 was assigned depending on their significance in WQ evaluation for human health (Solangi *et al* 2024).

5.2. Analytical Hierarchical Process

The Analytical Hierarchical Process is a popular analytical method that numerous researchers have adopted to effectively address complicated real-world decision-making approaches. This Knowledge-driven multi-criteria decision-making (MCDM) method is a prominent parameter weight prioritizing approach for contamination prediction (Kumar *et al* 2022). It works with three layers (target layer, criterion layer, and index layer) for efficiently solving complicated and interconnected problems (Chorol and Gupta 2023a, Cheng *et al* 2024). Experts are selected to score

the factors, and then the factors are compared. The score judgment matrix D is obtained, and then the eigenvectors of the judgment matrix ($W1$) are calculated by the geometric mean method (Li *et al* 2023). The advantage of the AHP method is that it simplifies decision-making by comparing pairs and synthesizing the outcomes (Kumar *et al* 2022). The authors employed AHP in the selected studies to identify groundwater-contaminated zones and assess groundwater vulnerability and GWQ. Sometimes, complex and uncertain environmental factors are addressed by incorporating fuzzy and stochastic uncertainty into the traditional analytical hierarchy process. Jing *et al* (2013) applied this hybrid methodology in water treatment technologies, demonstrating that the hybrid fuzzy stochastic AHP effectively addresses the uncertainty resulting from inadequate information and subjective biases in group decision-making scenarios, in contrast to the conventional AHP.

5.2.1. Groundwater vulnerability and contamination

AHP is widely used in several studies to assess groundwater vulnerability. Ojeda Olivares *et al* (2020) analysed to assess the impacts of anthropogenic and natural stresses on groundwater utilizing AHP and GIS methodologies. To comprehend the current and future spatial distributions of groundwater vulnerability in their model, they incorporated the climatological variables of temperature and precipitation. The vulnerability of the aquifer is highly characterized by abstraction, groundwater recharge and pollution influenced by changes in population and LULC (Land use and land cover change). A positive and immediate correlation was identified between pollution levels and socioeconomic marginalization, indicating that marginalized communities have higher water contamination levels due to inadequate wastewater treatment.

Several researchers employ the DRASTIC model in conjunction with AHP to determine groundwater vulnerability. In their study, Cheng *et al* (2024) utilized the AHP to assess the risk of groundwater pollution. They considered both the vulnerability of the groundwater and the contamination load from various sources. According to the adopted AHP-DRASTIC model, over 40% of sites are classified as being in high and relatively high sensitivity zones, indicating a greater susceptibility to fluoride contamination. Intense evaporation and prolonged natural water-rock contact increase fluoride concentration and salt deposition in groundwater. Another study conducted by Ozezin *et al* (2024) in Nigeria also assesses groundwater's susceptibility to contamination by combining DRASTIC-AHP models. Additionally, they include Land use (L) with the DRASTIC model to develop a DRASTIC-L-AHP model to evaluate potential contamination and risk. Results revealed that the DRASTIC-L-AHP model detected contaminated zones with the highest

accuracy, and sensitivity analyses showed that the vadose zone is the most efficient variable.

The Analytic Hierarchy Process is also used for heavy metal contamination and associated risk with geospatial modelling and Monte-Carlo simulation. Chorol and Gupta (2023a) utilized Pearson's correlation analysis and principal component analysis (PCA) to detect linear correlations and determine the sources of contamination in addition to AHP to assess heavy metal pollution. They analysed the contamination of Cr, Cd, Fe and Mn, and the result indicated that concentrations of Cr, Cd and Fe exceed the permissible values, and most of the samples fall under high pollution categories. Moreover, the sensitivity analysis showed that the metal concentration had the greatest impact on overall health risk, followed by body weight and exposure length. In a separate study, Chorol and Gupta (2023b) addressed a new combination of the analytic network process (ANP)—entropy and Holt's exponential smoothing (HES) model to predict GWQ, particularly for nitrate and fluoride in shorter periods with the associated health risks. ANP is a sophisticated technique for solving hierarchical, interdependent criteria and sub-criterion-based problems in decision-making. Additionally, the reason for selecting HES is that it adjusts changes in trends, such as WQ fluctuations across seasons for short time series without seasonal variations. They found that the Hybrid ANP-entropy model's prediction was more accurate and reliable in terms of uncertainties. In contrast, HES model provided unbiased results. The comprehensive PCA revealed that fluoride contamination resulted from the dissolution of minerals in felsic igneous rocks with dioritic compositions. Conversely, nitrate pollution resulted from the extensive use of fertilizers and the seepage from soak sites. However, considering the noncarcinogenic risks, fluoride poses greater risks than nitrate.

5.2.2. GWQ assessment

The relative importance of each parameter in predicting the GWQ parameters is very important. AHP and ANP (analytical network process) are widely utilized techniques for assessing a variable's relative significance or weight in a model. AHP is a decision-aided method, while ANP is the generalized form of the AHP method (Mokarram *et al* 2019). Some studies integrated different MCDM methods to overcome the problems associated with conventional approaches and obtain more reliable outcomes. Li *et al* (2023) evaluated the GWQ in a part of China using the coupled AHP-entropy weight-GRA method. The results revealed that the main factors affecting the GWQ in that area are TDS, TH, SO_4^{2-} , Cl^- , NO_3^- , and F^- , while the correlation analyses determined that anthropogenic activities contribute to pollution.

To overcome issues such as parameter selection, sampling, error in analyses and imbalance in the weighted parameters, Srivastava *et al* (2024)

conducted groundwater analyses on a shallow aquifer in India. They chose three decision-making methods—weighted sum model, GRA, and Technique for Order of Preference by Similarity to Ideal Solution—with two integrated weighted approaches (Distance-Relation method-GRM and Bayes method-BA) and AHP for effective decision-making. The highest consistency with GWQI ranking was found for DRM-GRA and EWM-GRA integration, indicating the MCDM method's potentiality in GWQ assessment. Considering the importance of groundwater contamination and risk analyses, Mokarram *et al* (2019) advised a GWQ model by employing the GIS-based analytical network process (ANP) and analytical hierarchical process (AHP). The fuzzy method is also applied to create a fuzzy map of the GWQ parameters using the trapezoidal membership function. The results showed that ANP achieves greater accuracy compared to fuzzy-AHP. Furthermore, it was shown that Ca, Cl, electrical conductivity (EC), and Na significantly influenced GWQ conditions.

5.3. GIS-based drastic model

Geospatial analysis techniques, such as GISs and remote sensing, provide valuable tools for mapping and analysing spatial data related to groundwater ecosystems. These techniques can identify vulnerable areas based on factors such as land use, soil characteristics, topography, and proximity to pollution sources. By integrating diverse datasets, geospatial analysis facilitates the identification of hotspots where climate change and contamination pose the greatest risk to GWQ and ecosystems. Vulnerability assessment of groundwater (GWVA) measures how sensitive groundwater is to its surrounding ecosystems. It is a useful tool for planning and deciding strategies for managing groundwater. GWVA is assessed using a variety of criteria, and the selection of these is contingent upon the characteristics of the research field and the available data (Abu-Bakr 2020). Seven parameters are used in the DRASTIC approach (Aller *et al* 1987) to determine the mapping of groundwater vulnerability: depth to groundwater, recharge, aquifer media, topography, impact of vadose zone, aquifer hydraulic conductivity, and texture type of soil media (Abu-Bakr 2020, Saranya and Saravanan 2023). The US Environmental Protection Agency (EPA) developed the DRASTIC method to map groundwater vulnerability using the seven hydrogeological criteria. The DRASTIC method was initially employed by Aller *et al* (1987) to evaluate vulnerability by evaluating rates and weights (Balaji *et al* 2021). In many circumstances, vulnerability mappings are used to assess contamination threats quickly. Therefore, the GIS-based qualitative techniques, which offer easier data accessibility, are affordable and require less time, are the most chosen tools for complex vulnerability assessment (Torkashvand *et al* 2021).

An accurate and effective investigation of groundwater susceptibility is critical for maintaining a satisfactory groundwater environment (Hui *et al* 2022). A significant amount of research has been conducted on groundwater vulnerability assessment by using the GIS-based DRASTIC model. More lower drastic index indicates the groundwater is more susceptible to pollution (Ahmed *et al* 2021). The GIS-based DRASTIC model is used by Ahmed *et al* (2021) to assess groundwater vulnerability in Pakistan. They found that almost 45% of the study area was in the moderately vulnerable zones according to the vulnerability classification. They also considered the spatial distribution of nitrate for the validation of DRASTIC indices, and the results demonstrated a good relationship between spatial nitrate distribution and groundwater vulnerability. However, Saranya and Saravanan (2023) utilized the DRASTIC model in a modified manner with AHP and evidential belief function (EBF) to develop the AHP-DRASTIC and EBF-DRASTIC groundwater vulnerability mapping methods. They developed the thematic map with seven DRASTIC parameters. EBF is a spatial integration model represented in mathematical terms with mass functions of a series of beliefs, disbelief, uncertainty, and plausibility for better assessment. The outcome of the analyses reveals that although AHP performed best in the multicriteria decision problem, the EBF's accuracy was higher, with an AUC value of 0.866. Furthermore, the EBF approach explicitly delineated the spatial correlation between the degree of vulnerability and each parametric class by calculating the Belief and Disbelief functions. In another study, Hui *et al* (2022) also modified the original DRASTIC model. They verified the effectiveness of the variable weight model. For their assessment, they replaced the topography, aquifer media, and impact of the vadose zone by land use type, aquifer thickness, and hydraulic resistance. They developed the AHP-DRASTIC, AHP-DRASHCL and VW-DRASHCL methods by optimising the weight using AHP and variable weight theory. The improved drastic approach VW-DRASHCL yielded the highest accuracy for the vulnerability index, suggesting that enhancing the parameters can improve the model's accuracy.

5.4. Physically based modelling

Hydrological models simulate water movement through the hydrological cycle, including infiltration, groundwater flow, and recharge processes. These models can be used to assess the impact of climate change on groundwater resources by simulating changes in precipitation patterns, temperature, and evapotranspiration rates. Additionally, hydrological models can incorporate contaminant transport algorithms to predict the spread of pollutants in groundwater systems.

Excessive groundwater withdrawal than natural or artificial recharge exacerbates the situation of water pollution, where groundwater simulation modelling serves as a valuable analytical approach. The correct determination of susceptible aquifers and the anticipation of potential pollution hazards rely on integrating factors, including hydrological variables, conditions in the climate, quantitative and qualitative water properties, the impact of vegetation, and other relevant variables. In this context, hydrogeologists and researchers employed MODFLOW, a popular hydrological model used in numerous studies (Nejatian *et al* 2024). Another hydrological model is the SWAT model, which is primarily employed to determine hydrological parameters, such as groundwater recharge, surface runoff, lateral flow, evapotranspiration, base flow, return flow, and soil water content (Bera *et al* 2024). Researchers have effectively employed these two models to assess the susceptibility and dynamics of groundwater contamination.

5.4.1. MODFLOW

MODFLOW is a groundwater-centric model that employs finite difference methods to resolve the mathematical equations that regulate three-dimensional groundwater flow developed by the United States Geological Survey (Nejatian *et al* 2024, Zhang *et al* 2024). This three-dimensional groundwater flow model is based on physical principles and is employed to address groundwater flow challenges in porous medium. The model has proven to be a powerful and versatile tool for simulating various aquifer parameters and their impacts (Nejatian *et al* 2024). MODFLOW was implemented by Nejatian *et al* (2024) and Zhang *et al* (2024) to evaluate groundwater susceptibility and identify potential pathways for groundwater contamination. Nejatian *et al* (2024) first use the DRASTIC method to assess the vulnerability and then use the long-time calibrated MODFLOW model to quantify aquifer dynamics. Apart from that, they also simulated pollutant transport through numerical MODPATH analyses. MODPATH analyses demonstrate that contaminants can migrate up to one kilometre during three years in the alluvial zone, with centre wells exhibiting greater barrier lengths than periphery wells. On the other hand, Zhang *et al* (2024) coupled MODFLOW with SWAT and 3D reaction rate model RT3D to simulate the nitrate fate and transport in a watershed of China. The RT3D model is mainly capable of simulating the advection, dispersion and chemical reaction of various chemical components of groundwater. Throughout the simulated period, nitrate interaction between surface and groundwater exhibited a positive correlation with precipitation, mostly driven by water exchange. Furthermore, to identify the vulnerable areas under changing climate scenarios and predict the impact of climate change on groundwater, Nassery and Salami (2016) used MODFLOW

along with climate and ML models. The outcome of the study showed that climate change will prolong groundwater depletion in the aquifer, and bedrock is mostly covered with marly formation, which will lower WQ if the groundwater head continues to drop.

5.4.2. SWAT

The SWAT model is another robust hydrological model which significantly predicts hydrological data in watershed modelling. It is primarily employed to determine hydrological parameters, such as groundwater recharge, surface runoff, lateral flow, evapotranspiration, base flow, return flow, and soil water content (Bera *et al* 2024). The model operates by initially dividing the watershed into smaller sub-watersheds or sub-basins and subdividing them into Hydrologic Response Units to enhance computing efficiency. Obtaining a more comprehensive understanding of the calibration parameters that describe catchment dynamics has the potential to result in an outstanding simulation output that exhibits reduced uncertainty in the model (Dash *et al* 2021).

To evaluate groundwater drought risk in the semi-arid Kansai River basin, Bera *et al* (2024) applied the SWAT model to analyse the hydrological budgetary component of the river basin, facilitating a more thorough investigation of water flow and other hydrological processes within each sub-region. Moreover, Zhang *et al* (2024) employed this model to simulate China's surface and groundwater nitrate-controlling factors and spatiotemporal variations. The findings indicated that grassland and agricultural land contributed higher nitrate levels in river water and groundwater than bare land and forest in the study area. Apart from that SWAT model is adopted for identifying the vulnerable areas under a changing climate scenario by Dash *et al* (2021). Using the calibrated and validated SWAT model for four various future climatic scenarios assuming unchanging land use, soil type, and agricultural management methods for the study basin, their assessments measured the effect of climate change on hydrology and stream-flow regime. The climate projections indicated the increase in minimum and maximum temperatures as well as precipitation. At the same time, the vulnerability assessment revealed an increase in moderate and high erosion-prone regions in the mid-century.

6. Assessment of GWQ with ML algorithms

In recent years, in addition to conventional analytical modelling approaches, data-driven methods such as ML have been increasingly popular for GWQ research. This is due to their capacity to predict complex and large datasets (Habib *et al* 2023, Salaudiddin *et al* 2023), as well as to handle missing data for the analysis and forecasting of WQ (Alshehri and Rahman 2023). Several studies have documented the

application of ML algorithms in water resource management, particularly in groundwater contamination (Mosavi *et al* 2021, Pham *et al* 2022), GWL predictions (Rohde *et al* 2021), and (GWQ estimations (Lap *et al* 2023, Sahour *et al* 2023) worldwide. Additionally, previous studies have also focused on assessing individual parameters of GWQ, such as nitrate (Sarkar *et al* 2022), arsenic (Tan *et al* 2020), fluoride (Araya *et al* 2022), and the combined effects of heavy metals (Huynh *et al* 2022). Several recent investigations have focussed mainly on nitrate contamination using various ML algorithms (Band *et al* 2020, Pham *et al* 2022, Liu *et al* 2024). Overall, analysing and predicting GWQ parameters using an ML algorithm involves several stages, starting with required data collection and selecting accurate parameters for analysis and forecasting. The prediction of GWQ using an ML algorithm typically involves the following steps:

- i. Collection of groundwater samples,
- ii. Computation of WQI
- iii. Develop, train and test ML-based prediction models
- iv. Computation of error measures (MSE, MAE, RMSE, RSR, NSE, MAPE, R^2 , etc) and
- v. Recommendation regarding the performance of ML algorithms.

6.1. Sample collection timescales and analyses in each publication

The first and most important step in analysing and predicting GWQ using a ML technique is identifying the most relevant GWQ parameters for the particular catchment and their standard concentrations/thresholds. Table 4 presents the most analysed GWQ parameters in the reviewed literature and their standard threshold values or concentrations. It is evident from table 4 that TDS (27) and EC (26) were examined in most studies, whereas *Escherichia coli* (*E. coli*) was examined in only three papers, the majority of which were conducted in Ireland. Furthermore, SO_4^{2-} , Na^+ , Ca^{2+} , Mg^{2+} , Cl^- , NO_3^- , and HCO_3^- were the significant parameters included in the analysis for GWQ forecasts. Moreover, a few studies (table 5) investigated other quality parameters (TSS, alkalinity, Fluoride, Mn, Zn, Cr, Cd) to evaluate their impact on groundwater.

Sixty-three distinct types of input parameters were employed across all research papers. Figure 3 illustrates the input parameters most frequently analysed by more than five studies. Among these, 16 parameters were frequently used by different researchers, as shown in figure 3. The most commonly used parameter was TDS, which was analysed in 27 publications. In contrast, the least analysed were Mn and turbidity, assessed in 5 publications each. Following TDS, EC was the second most analysed parameter (26). Additionally, pH, Na, Ca, Mg, and SO_4^{2-} were also widely used parameters for GWQ

analyses and were assessed in 23 different studies. Additionally, chloride, potassium, bicarbonate, total hardness, nitrate, and iron were mentioned in more than 10 articles.

Figure 4 represents the timeline for data collection in the selected studies. As highlighted in figure 4, most studies (>20) did not report the timeframe for data collection for analyses, where multiple datasets were used. The most commonly reported data collection timescale was yearly, ranging from one year to more than 30 years for the prediction of GWQ. Additionally, seasonal (such as dry and wet seasons), irregular, and single-month data collection were also reported in a few studies.

6.2. Computation of GWQI

The relative classification of the individual parameters is determined by calculating the GWQI. The majority of the research followed the widely used weighted arithmetic method to compute the WQI. The range of classification of water in different countries is nearly identical. The following tables (tables 6 and 7) show the steps of the calculation of the WQI and the water classes employed by researchers in different countries.

6.3. ML-based prediction algorithms

6.3.1. SVM

SVM is a supervised ML algorithm that is often used for classification problems and solving high-dimensional and nonlinear pattern recognition (Abu El-Magd *et al* 2023, Lap *et al* 2023, Sahour *et al* 2023). In addition to classification problems, recent research has highlighted the application of the SVM algorithm for regression analysis, including the prediction of GWQ parameters (Lap *et al* 2023, Sahour *et al* 2023). The SVM algorithm was originally developed for binary classification, using hyperplanes to identify decision boundaries between data points of different classes (Abu El-Magd *et al* 2023). In SVM, the data is divided into regressive subclasses using a decision surface called a hyperplane, which maps the non-linear input space onto a higher-dimensional region. The performance of SVM is determined by the kernel function used, with common functions including linear, polynomial, radial basis function, and sigmoid kernel functions (Shiri *et al* 2021).

Several recent studies (Aish *et al* 2023, Lap *et al* 2023, Nasiri Khiavi *et al* 2023, Sajib *et al* 2023) have reported the application of the SVM algorithm in estimating GWQ parameters for drinking and irrigation purposes in arid regions, as well as nitrate and saline contamination in groundwater wells. This ML algorithm proved to be particularly effective in predicting irrigation WQ and saline contamination. For example, Ibrahim *et al* (2023) employed both an ANFIS and SVM techniques to assess GWQ for irrigation purposes. They reported that while both models showed reasonable accuracy in simulating the

Table 4. Frequently analysed GWQ parameters in the reviewed literature, along with their standard threshold values or concentrations.

Parameters	Analysed in the number of studies	Threshold concentrations WHO	Ireland (WFD Standard), Irish threshold	BIS (Bureau of Indian Standards)	GSA—Ghana Standard (Apogba <i>et al</i> 2024)		
EC ($\mu\text{S cm}^{-1}$)	26	1500	Sajib <i>et al</i> (2023)	—	—	8.5	
pH	24	6.5–8.5	Nourani <i>et al</i> (2024)	—	6.5–8.5	Venkatesh <i>et al</i> (2024)	1000
SO ₄ ^{2−} (Mg/l)	23	250	Nourani <i>et al</i> (2024)	—	200–400	Raheja <i>et al</i> 2024	250
Na+ (Mg/l)	23	200	Nourani <i>et al</i> (2024)	—	20–200	Venkatesh <i>et al</i> (2024)	—
NO ₃ [−] (Mg/l)	16	45	Sunita and Ghosh (2024)	37.5	45	Raheja <i>et al</i> (2024)	50
Ca ²⁺ (Mg/l)	23	200	Ben Brahim <i>et al</i> (2021)	—	75–200	Raheja <i>et al</i> (2024)	75
Mg ²⁺ (Mg/l)	23	30	Nourani <i>et al</i> (2024)	—	30–100	Raheja <i>et al</i> (2024)	50
TDS(Mg/l)	27	1000	Nourani <i>et al</i> (2024)	—	500–1500	Raheja <i>et al</i> (2024)	500
HCO ₃ [−] (Mg/l)	16	120	Nourani <i>et al</i> (2024)	—	—	Venkatesh <i>et al</i> (2024)	—
K ⁺ (Mg/l)	20	12	Nourani <i>et al</i> (2024)	—	12–50	Venkatesh <i>et al</i> (2024)	—
Cl [−] (Mg/l)	22	250	Nourani <i>et al</i> (2024)	—	250–1000	Raheja <i>et al</i> (2024)	250
TH(Mg/l)	12	500	Nourani <i>et al</i> (2024)	—	200–600	Raheja <i>et al</i> (2024)	500
Fe (Mg/l)	10	0.3–1	Sajib <i>et al</i> (2023)	—	—	—	0.3
P(Mg/l)	—	—	—	0.035	—	—	—
<i>E. Coli</i>	3	0/100 ml	—	—	—	—	—

Table 5. Other GWQ parameters used for analyses in the selected papers.

Other parameters	Analysed in the number of studies	Threshold concentrations (mg l ^{−1}), India, WHO, BIS (Bureau of Indian Standards), ECR	References
TSS	2	500 (BIS)	Jha <i>et al</i> (2020)
Fluoride	4	1 (BIS)	Jha <i>et al</i> (2020)
Alkalinity	2	200 (BIS)	Sunita and Ghosh (2024)
Mn	2	0.4 mg l ^{−1} (WHO 2011)	(El-Rawy <i>et al</i> 2024; Sajib <i>et al</i> 2023)
Zn	1	5 mg l ^{−1} (ECR 2023)	Sajib <i>et al</i> (2023)
Cr	2	0.05 mg l ^{−1} (ECR 2023)	Sajib <i>et al</i> (2023)
Cd	2	0.003 mg l ^{−1} (ECR 2023)	Sajib <i>et al</i> (2023)
Cu	1	1.5 mg l ^{−1} (ECR 2023)	Sajib <i>et al</i> (2023)

irrigation WQI, the SVM outperformed the ANFIS model, achieving a determination coefficient (R^2) of 0.97 for training and 0.70 for testing. Furthermore, Mosavi *et al* (2021) argued that SVM models outperformed other predictive models, such as RF, in groundwater salinity mapping.

6.3.2. RF

RF is a popular bagging-based DT algorithm often used to address multi-classification and regression problems (El-Rawy *et al* 2024, Thanh *et al* 2024). It is an ensemble learning method in which a randomly selected subset of all features is used to create an

ensemble of DTs. The final predicted quantity is calculated based on the predictions made by each DT in the ensemble, and the best predictor model is used for this estimation (Habib *et al* 2023, Salauddin *et al* 2023). Numerous research in recent years has been devoted to the application of the RF algorithm for predicting GWQ and contamination, viz., nitrate and salinity (Singha *et al* 2021, Senapati *et al* 2024, Solangi *et al* 2024, Venkatesh *et al* 2024). In this review, over 21 studies employed the RF ML algorithm to predict GWQ, with RF identified as the best model in six studies. For instance, a recent work by Thanh *et al* (2024) explored the use of RF algorithms with other

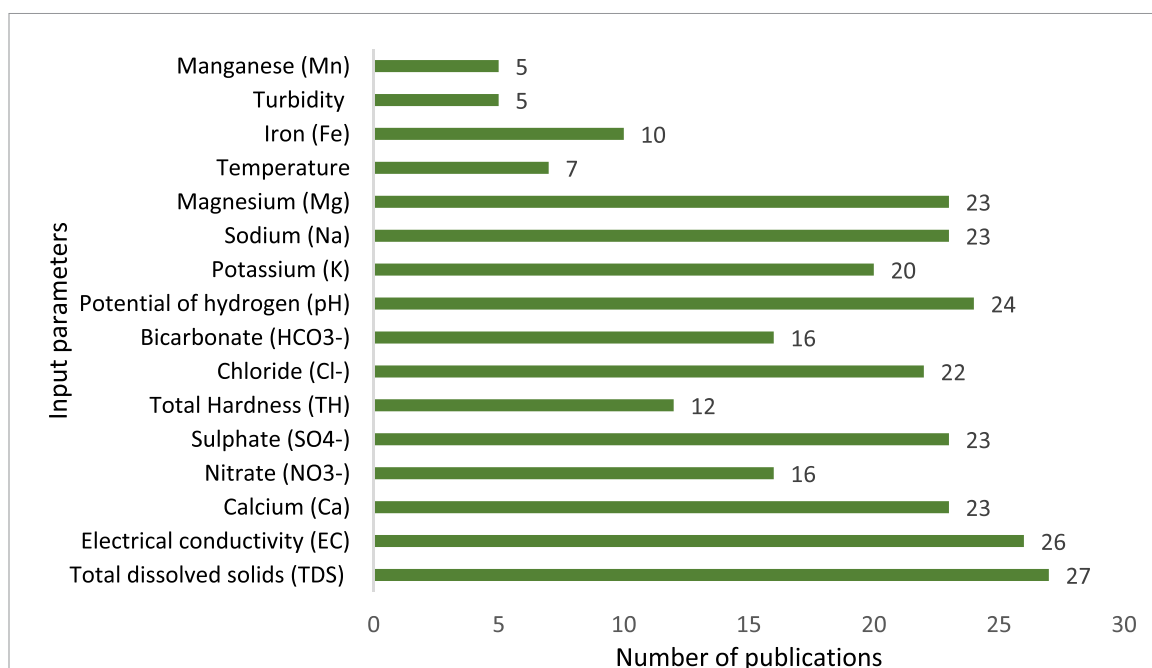


Figure 3. Graphical analysis of input parameters >5 in the studied papers.

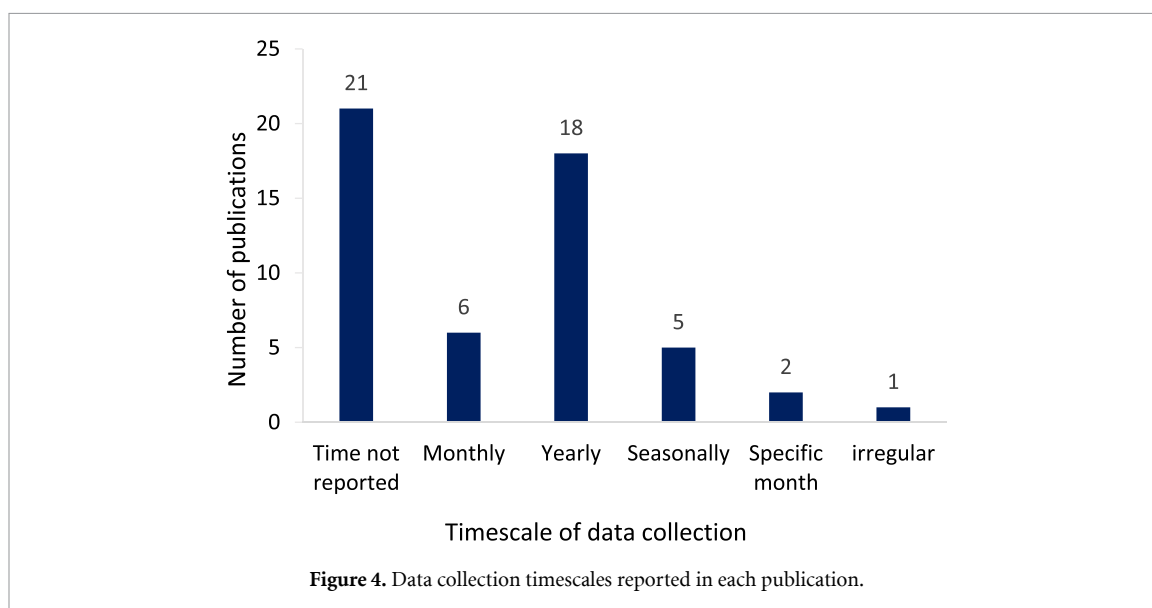


Figure 4. Data collection timescales reported in each publication.

ML models to predict GWQ. The authors found that RF outperformed the other tested ML models for the prediction of GWQ. Similar findings were also reported in several studies regarding the prediction of groundwater nitrate concentration. For example, Band *et al* (2020) and Sarkar *et al* (2022) the RF model performed better than others in predicting groundwater nitrate contamination. Apart from that, RF also provided the best performance when embedded with the sequential forward floating selection, as indicated by a recent study conducted by Rodriguez-Galiano *et al* (2018).

Additionally, Rohde *et al* (2021) employed a RF algorithm to predict GWLs in GDEs across California, USA. The authors highlighted several benefits of the

RF algorithm, such as its computational efficiency, ability to avoid overfitting, and capability to handle numerous predictors, including continuous, categorical, or nominal (binary) data, even from small datasets with observations.

6.3.3. Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is a supervised ML algorithm known for its exceptional accuracy, efficiency, and versatility. It is a cornerstone in predictive modelling (El-Rawy *et al* 2024). XGBoost is highly adaptable and capable of handling various regression and classification tasks. It utilizes the Classification and Regression Trees (CART) framework as its base classifier, creating an ensemble of

Table 6. Estimation of the water quality index.

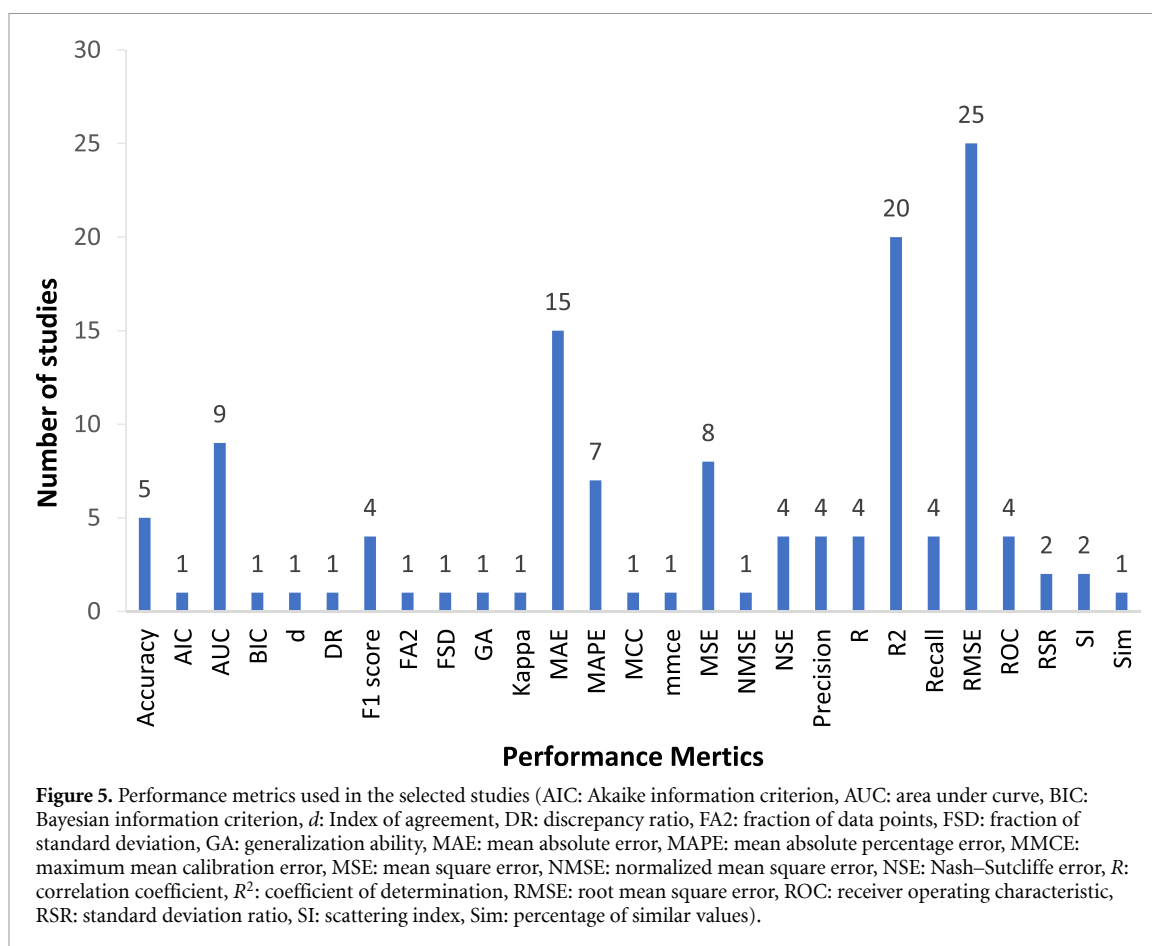
Steps	Equations
Step 1: Estimate the relative weight (W_i)	$W_i = \frac{w_i}{\sum_{i=0}^n w_i}$ where w_i is the weight of each parameter and n is the number of parameters.
Step 2: Calculate the quality rating (q_i)	$q_i = \frac{C_i}{S_i} \times 100$ where C_i is the concentration of chemicals in the water sample in (mg l^{-1}) and S_i is the drinking water standard for each chemical parameter.
Step 3: Determining the subindex value (SI_i)	$SI_i = W_i \times q_i$ where, W_i is the relative weight and q_i is the quality rating.
Step 4: The overall WQI is calculated by aggregating the sub-indices.	$WQI = \sum_{i=0}^n SI_i.$

Table 7. Water classification appeared in studies and legislations.

References	Country	WQI values/GWQ score range	Water classification
Lap <i>et al</i> (2023)	Vietnam	91–100	Excellent
		76–90	Good
		51–75	Medium
		26–50	Bad
		<25	Very bad
Solangi <i>et al</i> (2024)	Pakistan	Below 50	Excellent water
		50–100	Good water
		>100–200	Poor water
		>200–300	Very poor
		>300	Unfit for drinking
Sahour <i>et al</i> 2023,	Iran	0–25	Excellent
		26–50	Good
		51–75	Poor
		>75	Very poor
Sajib <i>et al</i> (2023), Uddin <i>et al</i> (2024)	Bangladesh	80–100	Good
		50–79	Fair
		30–49	Marginal
		0–29	Poor
Elmeddahi and Ragab (2022)	Algeria	<50	Excellent
		50–100	Good
		100.1–200	Poor
		200.1–300	Very poor
		>300	Unsuitable for drinking

interconnected DTs. Each subsequent tree uses input samples informed by the training and predictions of previous trees. In this review, XGBoost was found to be one of the most commonly employed ML models for predicting GWQ, such as Krishnamoorthy and Lakshmanan (2024), Hussein *et al* (2024), Raheja *et al* (2022), Bedi *et al* (2020). For instance, Bedi *et al* (2020) concluded that XGBoost, being a tree-based model, facilitates model interpretability, which makes it a good alternative when compared to other approaches for predicting WQ. Furthermore, several studies have identified the XGBoost ML algorithm as the best-performing predictive model for their research. For example, Tran *et al* (2021) reported

that the XGBoost ML algorithm performed better compared to other tested ML models for the identification of the saline contamination in the coastal multi-layer aquifers of the Mekong River Delta of Vietnam. In a recent study by Solangi *et al* (2024), it has been claimed that up to 100% accurate prediction of the WQI can be achieved by employing an XGBoost algorithm. Furthermore, the study by Zheng *et al* (2024) also reveals that XGBoost outperforms the DT and RF models in predicting WQ. In addition to a plain XGBoost algorithm, previous studies have also employed XGBoost-based SHAP (SHapley Additive exPlanations) values to improve decision-making strategies in water management.



6.3.4. Other ML models

In addition to SVM, RF, and XGBoosting, previous studies also reported the application of other supervised and unsupervised ML algorithms, including ANNs, k-nearest neighbour (KNN), multiple linear regression (MLR), ridge regression (RR), and Gaussian classifier model (GCM) for GWQ predictions. For example, Sahour *et al* (2023) utilized several ML algorithms in addition to SVM, RF, and XGBoost, such as ANNs, KNN, and GCM for predicting GWQ. They found that RF yielded the best prediction performance for their case compared to other algorithms. Furthermore, another study by Shiri *et al* (2021) used a modified hybrid version of the SVM algorithm with firefly algorithm (FA) to simulate groundwater wells for four key WQ variables: EC, sodium adsorption ratio (SAR), total dissolved solids (TDS), and sulphate (SO_4^{2-}). This approach demonstrated the best performance. Section 6.5 provides an overview of the employed ML algorithms and their performance.

6.4. Performance metrics for predicting GWQ

Performance evaluation matrices are quantitative measures used to assess the accuracy and effectiveness of developed predictive models (Huang *et al* 2024). The evaluation matrices such as coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), Mean absolute percentage

error (MAPE), mean squared error (MSE), normalized mean square error (NMSE), root mean squared error (RMSE) play an important role in assessing the performance of predictive models. R^2 in this case, represents the proportion of variance explained by the model, and RMSE and MSE quantify prediction accuracy. These metrics facilitate model comparison, the selection of the most high-performing one, and ensure reliability in various applications (Nourani *et al* 2024).

Figure 5 illustrates the most frequently used performance metrics for analysing different types of ML models. RMSE is the most commonly used metric for evaluating performance, as reported by 25 studies, followed by R^2 , which was used in 20 studies. In total, various studies employed nearly 30 different performance metrics to assess the models' performance. Additionally, most studies used more than one metric to evaluate prediction performance. However, performance metrics in ML can sometimes make it challenging to identify the best-performing model, as different metrics may highlight varying aspects of model performance.

6.5. Best-performing ML algorithms for predicting GWQ

Over the past decade, various ML algorithms have rapidly developed and applied in water and environmental engineering research, particularly for

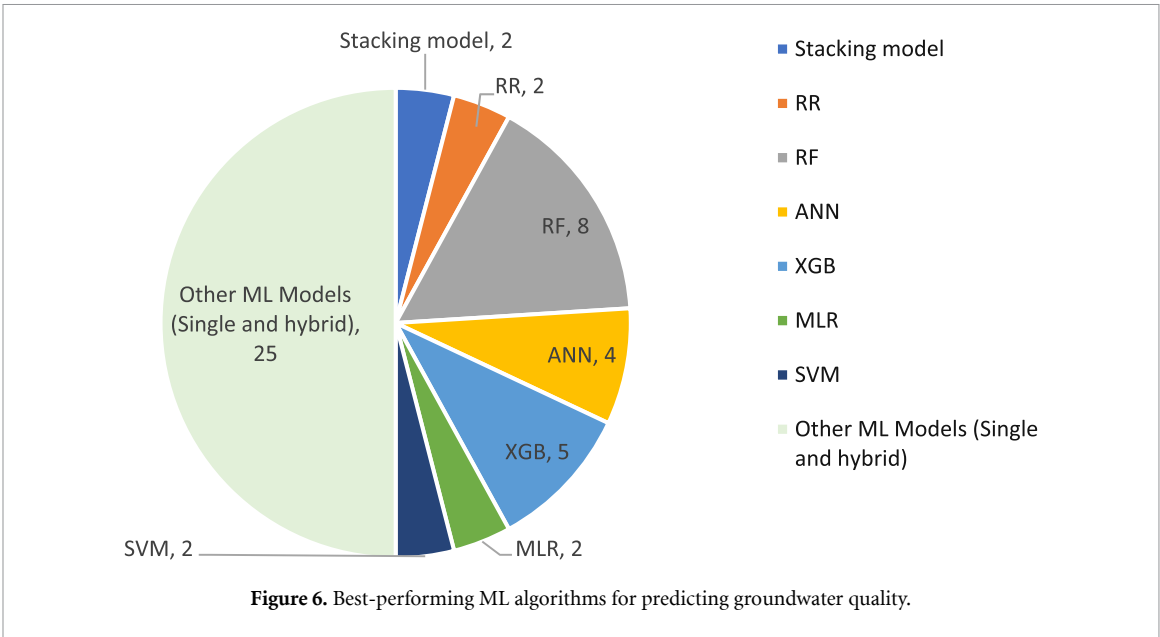


Table 8. Overview of machine learning techniques applied to groundwater quality prediction.

ML technique	Application	Groundwater parameters	Advantages	Challenges	References
Random Forest	Predicts contamination events	pH, Heavy metals, Nitrates	Handles high-dimensional data	Requires large datasets	(Makhlouf <i>et al</i> 2024; Sahour <i>et al</i> 2023; Shiri <i>et al</i> 2021)
Support vector machines	Classifies groundwater quality levels	Salinity, organic pollutants	Effective with small datasets	Sensitive to parameter tuning	(Bedi <i>et al</i> 2020; Lap <i>et al</i> 2023; Shiri <i>et al</i> 2021)
Artificial neural networks	Forecasts changes in water quality over time	Nitrate, temperature, pH	Handles non-linear relationships	Overfitting requires large training data	(Bedi <i>et al</i> 2020; Kouadri <i>et al</i> 2022)
Decision Trees	Detects influence of climate variables	pH, total dissolved solids	Easy to interpret	Prone to overfitting without pruning	(Krishnamoorthy and Lakshmanan 2024; Lap <i>et al</i> 2023; Zheng <i>et al</i> 2024)
k-nearest neighbours	Identifies contamination sources	Contaminant levels (e.g. Arsenic)	Simple and intuitive	Computationally expensive for large datasets	(El-Rawy <i>et al</i> 2024; Rahmati <i>et al</i> 2019)
Gradient boosting machines	Predicts groundwater quality trends	Chloride, Sulphate	High accuracy	Complex and harder to interpret	(Alshehri and Rahman 2023; Sahour <i>et al</i> 2023; Sunita and Ghosh 2024)

predicting GWQ. The best-performing models, as reported in the reviewed literature, are shown in figure 6. It is clear that most studies have identified RF as the top ML algorithm for predicting GWQ parameters, followed by XGBoost. This modified DT algorithm provided the best results in 5 studies. Stacking, RR, and MLR models also showed the best performance in 2 of the reviewed research papers. Table 8 overviews the most popular and widely employed ML models for GWQ prediction and their relative advantages and challenges. Based on table 8, it is clear that each machine-learning

algorithm presents unique advantages and challenges. It is crucial to carefully weigh these factors when developing and implementing such algorithms for GWQ predictions.

7. Integration of analytical models with ML approaches in groundwater systems

Using analytical modelling in conjunction with ML techniques can help understand the complex interactions by evaluating the impact of climate change

Table 9. Comparative performance of analytical models vs machine learning techniques in predicting groundwater quality.

Criteria	Analytical models	Machine learning techniques	References
Data Requirements	High (Requires detailed physical data)	Moderate to High (Requires large and diverse datasets)	(Akakuru <i>et al</i> 2023, Bera <i>et al</i> 2024)
Accuracy	High in simulating physical processes	High in capturing non-linear relationships	(Akakuru <i>et al</i> 2023, Makhoulouf <i>et al</i> 2024)
Computational efficiency	Lower, time-intensive	Higher, faster once trained	(Ozegin <i>et al</i> 2024, Zhang <i>et al</i> 2024)
Interpretability	High, transparent processes	Lower, often black-box models	(Chorol and Gupta 2023b, Sahour <i>et al</i> 2023)
Adaptability to new data	Moderate, requires re-calibration	High, can learn from new data	(Nejatian <i>et al</i> 2024, Shiri <i>et al</i> 2021)
Key applications	Provide well-informed decision	Faster prediction	(Abu El-Magd <i>et al</i> 2023, Srivastava <i>et al</i> 2024)
Computation time	Time-consuming	Require less time	(Elmeddahi and Ragab 2022, Krishnamoorthy and Lakshmanan 2024)

on groundwater resources. By integrating data on GWLs, recharge rates, evapotranspiration, and land use changes, these models can offer insights into how climate-induced changes affect both the quantity and quality of groundwater (Dimitriadou and Nikolakopoulos 2021, Chen *et al* 2023). Additionally, modelling the vulnerability of groundwater to pollution under potential future climate change scenarios can aid in assessing the risks to GWQ (Li and Merchant 2013).

Analytical models, such as mathematical equations describing groundwater flow and solute transport, have long been employed to simulate hydrological processes and assess GWQ (Li *et al* 2023). However, traditional analytical models often struggle to capture the complexity and nonlinearity of real-world processes, particularly under changing climatic conditions. ML techniques, including ANNs, SVMs, and RF algorithms, have demonstrated remarkable capabilities in capturing complex patterns and relationships within large datasets. By integrating ML approaches with analytical models, researchers can augment the predictive capabilities of traditional models and improve their ability to account for nonlinearities and uncertainties in groundwater systems. ML techniques enable data-driven insights into the interactions between climate variables, land use changes, and GWQ parameters, facilitating more accurate predictions and scenario analyses.

Researchers employed hybrid machine-learning models to enhance the accuracy and reliability of groundwater monitoring and assessment. Shiri *et al* (2021) adopted an integrative FA in the multi-layer perceptron (MLP) and SVM models to create the integrative models of MLP-FA and SVM-FA. They employed the FA algorithm to update the tuning parameters of the standard SVM model including the hyperplane parameters, as well as weights and biases in the MLP neural network. Their analyses

revealed that the developed hybrid SVM FA (SVM-FFA) model achieved the best predictability performance. Additionally, Thanh *et al* (2024) integrated the RF and ANN models into cross-validation (CV) and bootstrapping (B) techniques. The result of their study indicated that GWQ in the study area was predominantly polluted by calcium, magnesium, and bicarbonate and that the RF-CV model outperformed the RF-B model.

Few studies have been conducted to predict the impact on groundwater by integrating analytical and ML modelling with their comparative and potential performance in decision-making and prediction. Both models are proficient in evaluating groundwater contamination and forecasting quality parameters; however, AM models excel in decision-making analysis, while ML is superior for predictions involving extensive datasets (table 9). Makhoulouf *et al* (2024) recently integrated the widely-used MODFLOW and ML models to simulate present and anticipated climate scenarios over 80 years from 2020 to 2100. MODFLOW results indicated a decline in GWLs due to prevailing climate conditions and the RCP 4.5 and RCP 8.5 scenarios projected for 2100. The calibrated MODFLOW data was subsequently utilized to construct the ML model employing four prominent algorithms: GPR, Support Vector Regression (SVR), DT, and ANN. GPR exhibited the highest accuracy in predicting pumping rates among all the models, with a correlation coefficient of 1.0. Similarly, Jose and Yasala (2024) integrated the hidden Markov model (HMM) with the ANN, creating a ML-based ensemble model to forecast GWQ in a district of Tamil Nadu, India. The combined approach of HMM-ANN demonstrated its effectiveness in GWQ prediction with an accuracy of 97.41%. A separate study (Bera *et al* 2024) in India employed analytical and ML modelling to evaluate groundwater drought, which jeopardizes water availability and negatively affects the groundwater ecosystem. The hydrological

modelling of SWAT and three distinct ML models of SVM, RF, and neural networks (NN) were amalgamated to evaluate the risk. The analyses demonstrated that the upper catchment areas are the primary location of high-risk zones. In contrast, the lower catchment regions are the location of lower-risk zones. In terms of performance, RF emerged as the best model, with an AUC-ROC accuracy of 91%.

A distinctive hybrid modelling approach was implemented in the groundwater of coastal regions of Bangladesh by (Uddin *et al* 2024). This was followed by the integration of the extreme gradient boosting (XGBoost) ML method with root mean squared (RMS) and WQI modelling. The RMS-WQI model demonstrated minimal uncertainty in predicting WQI scores, which was less than 1%. Additionally, XGBoost exhibited an R^2 value of 0.97, indicating exceptional sensitivity in accurately predicting GWQ. Table 9 presents the comparative performance of analytical and ML models for data analysis and management in the broad spectrum of groundwater research.

8. Knowledge gaps and future research priorities

This systematic review highlights the growing body of research on climate change's impacts on GWQ and the significant role of analytical models and ML techniques in understanding and predicting these changes. Nonetheless, there remain deficiencies and opportunities for further research regarding the long-term effects of climate change, the identification of sensitive regions, and the prediction of GWQ through individual analytical methods and ML, as well as their integrated performance.

The precipitation pattern, sea level rise, temperature increase, saltwater intrusion, and coastal inundation are all indicators of climate change's effects. In the past, research has concentrated on various aspects of climate change, such as its influence on groundwater recharge, the response of groundwater temperature to climate change, and the impact on groundwater microorganisms. While atmospheric temperature shifts are clearly linked to groundwater temperature changes, the impact of other climate-related factors, such as precipitation or land use changes, is not fully explored. Research should enhance subsurface heat transport models in cold and temperate locations by incorporating freezing, thawing, and heat advection via groundwater movement. Microbial and chemical contamination is another contributing factor in GWQ, and several studies have covered this topic extensively. While some studies show that microbial communities adapt rapidly to contamination, there is limited understanding of how these communities function during both the initial response and the

long-term recovery period. Therefore, future research should focus on the functional aspects of microbial communities and their abilities to resist pathogen invasion during the contamination and recovery phases. This could include studying key microbial processes, such as organic matter degradation and nutrient cycling, to understand the broader impacts of contamination on groundwater ecosystem functions. Additionally, nitrate contamination receives more attention than other GWQ indicators and is a global issue, especially in Ireland. Consequently, precise forecasts of nitrate pollution should be further prioritized using advanced ML models to facilitate the development of initiatives to improve GWQ. Moreover, future research should focus on developing integrated models that link groundwater flow, contaminant transport, and climate change scenarios to predict the spread of pollutants better.

Analytical models are widely used in GWQ and contamination analyses. The multicriteria decision-based AHP is the most widely used model. Other popular analytical models are the GIS-based DRASTIC model for vulnerability analyses and MODFLOW and SWAT for hydrological analyses. Identifying the contaminant source is one of the major steps in managing groundwater resources; however, there is limited understanding of the specific sources of both natural and anthropogenic contaminants affecting the aquifers through conventional modelling. Future research should focus on identifying specific sources of contaminants and evaluating methods to prevent or mitigate their impact on aquifers. Apart from that, individual studies concentrate solely on the effect of metals rather than considering the potential positive or negative interactions between metals in the groundwater. Further research should explore how multiple contaminants may interact and collectively affect health risks. Moreover, as an analytical approach, the fuzzy logic framework effectively handles nonlinearity and uncertainty; there may be room for improvement in refining the fuzzy membership functions and decision rules, especially for more complex, large-scale, or heavily polluted groundwater systems.

In recent years, researchers have focused on ML to improve the accuracy of GWQ predictions and to handle large datasets. The ML model has mitigated the risk of inaccurate measurements of the WQ parameters that could potentially mislead the prediction. Prediction with ML requires robust datasets; however, current research is deficient in data and primarily relies on historical data. Future models need to incorporate climate change estimates to enhance the precision and resilience of predictions amid extreme changing conditions. Furthermore, to improve prediction reliability, research should

Table 10. Future research directions in groundwater quality modelling under climate change.

Research area	Knowledge gaps	Suggested approaches	References
Integration of climate, contaminant transport and groundwater models	Lack of integration between models	Develop coupled models for holistic analysis	(Gintamo <i>et al</i> 2021, Kløve <i>et al</i> 2014, Makhoul <i>et al</i> 2024)
ML techniques for groundwater prediction	Limited dataset size and quality	Collect higher-resolution climate and GWQ data	(Sahour <i>et al</i> 2023, Shiri <i>et al</i> 2021, El Bilali <i>et al</i> 2021)
Impact of climate extremes	Insufficient focus on extreme weather impacts	Simulate groundwater behaviour under extreme events	(Egidio <i>et al</i> 2022, Kløve <i>et al</i> 2014)
Cross-regional analysis	Limited studies comparing different regions	Conduct cross-regional comparative studies	(Kurylyk and Macquarrie 2014, Jha <i>et al</i> 2020, Uddin <i>et al</i> 2024)

concentrate on a more thorough examination of uncertainties in the input data, models, and parameters. Another challenge in ML is managing imbalanced data, which necessitates the exploration of sophisticated hyperparameter tuning, particularly for ANNs, that could substantially improve prediction performance.

Similarly, a key area of future research combines several modelling techniques, including ML and analytical modelling, to anticipate groundwater contamination and identify susceptible zones. A limited number of studies have been conducted on GWQ and ecosystem management through hybrid modelling, facilitating the development of new and innovative research in this domain. There is a lack of integration and comparison in advanced modelling; hence, combining traditional and ML techniques, such as geostatistics and deep learning, may offer a more comprehensive approach to GWQ monitoring and management. This ensemble technique allows combining two or more ML algorithms to improve the predictive performance. Apart from that, current research predominantly concentrates on a localized or highly restricted region. Consequently, future hybrid models must incorporate GWQ and the global climate model on an expansive scale, rendering them versatile across interdisciplinary domains.

However, analysing dynamic climatic factors and GWQ parameters using hybrid modelling research incorporates a variety of diverse datasets. These datasets typically encompass climatic data (e.g., precipitation, temperature), hydrochemical data including TDS, NO_3^- , Na^+ , Cl^- , and geospatial data to map and analyse the geographic variability of GWQ over designated locations according to computational needs. The integration ensures comprehensive modelling by combining the spatiotemporal variability of GWQ with WQ standards. Furthermore, it ensures a robust framework that identifies temporal patterns through models like the HMM and non-linear correlations via ML (e.g., ANN), hence improving prediction reliability. Nevertheless, the proposed hybrid framework introduces additional

computational requirements compared to conventional models. A hybrid fuzzy-GIS-based GWQI, for example, classifies parameters into qualitative categories such as ‘desirable,’ ‘acceptable,’ and ‘unacceptable’ based on WQ requirements. This step demands significant computational resources to handle huge datasets and ensure classification consistency. Apart from that, long-term (80 year) simulations under multiple scenarios (e.g., RCP 4.5 and RCP 8.5) sometimes required substantial computational power to model groundwater dynamics and climate impacts on aquifers. However, incorporating ML reduces errors while considerably improving prediction reliability and accuracy, providing decision-makers with actionable insights. A comprehensive summary of existing knowledge gaps and future suggestions are presented in table 10 and figure 7.

9. Summary

The primary objective of this paper was to systematically review and synthesize existing literature on the impacts of climate change on GWQ and the application of analytical and ML modelling approaches to investigate how climate change impacts GWQ. Additionally, the critical research gaps are identified, and an outlook on future research direction is provided. Eighty-nine relevant studies were identified based on comprehensive keyword retrieval combined with Analytical and ML modelling. Key data and relevant information from the selected papers were compiled and used to perform a systematic literature analysis. This review outlines the intricate interactions between climate-induced variables, such as changes in precipitation regimes, fluctuations in temperature, sea level rise, and extreme hydro-climatic events, and their effects on GWQ. These factors induce a cascade of geochemical, microbial, and physical processes that degrade groundwater resources, with implications for primary productivity, microbial activity, and community health. In addition to analytical models, over the past decade, the use of ML in GWQ modelling has expanded rapidly, with

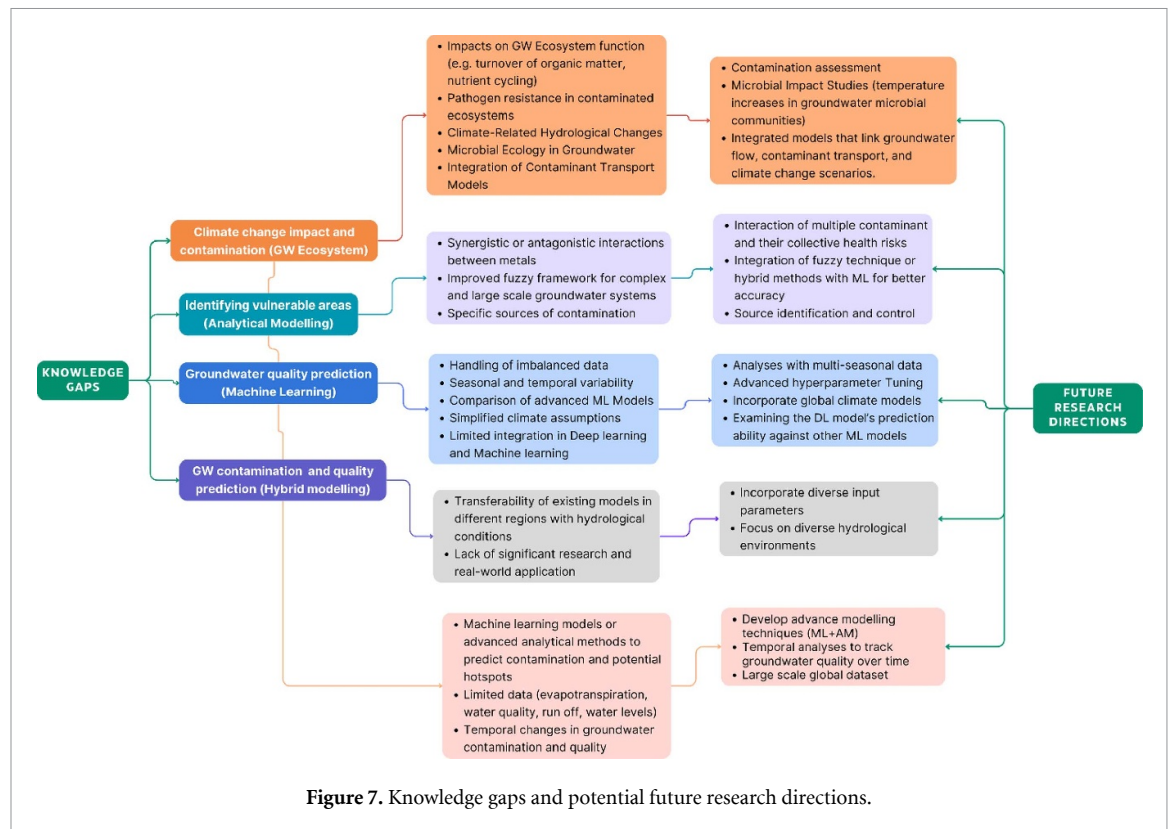


Figure 7. Knowledge gaps and potential future research directions.

increasing emphasis on hybrid models that integrate ML with traditional analytical approaches.

The multilevel meta-analysis highlights the critical role of hybrid modelling by combining ML and analytical models in advancing our understanding of these complex dynamics. These models offer robust frameworks for GWQ potential assessments, spatial mapping, and threshold estimation, facilitating the prediction and management of GWQ under varying climatic scenarios. The integration of SWAT-derived hydrological parameters with ML models facilitates a thorough assessment by utilising both deterministic and data-driven approaches. The SWAT model effectively integrates climatic variability, such as alterations in precipitation and evapotranspiration, into hydrological simulations, making it relevant in dynamic climate scenarios. The statistical analysis (e.g. Mann–Kendall Test with Sen's Slope) validates the spatial and temporal trends in GWLs, confirming the reliability of the models in identifying high groundwater drought risk (GWDR) zones. Furthermore, the AUC-ROC performance curve offers a quantitative assessment of the predictive accuracy of the ML model, with RF (91%) surpassing other models (SVM, NN) (Bera *et al* 2024). Nonetheless, an alternative hybrid modelling analysis (MODFLOW and ML) under current climate scenarios (RCP 4.5 and RCP 8.5) revealed that the GPR exhibited superior performance, achieving a correlation coefficient of 1 (Makhlouf *et al* 2024).

Moreover, integrating ML with traditional analytical models remains a promising frontier in GWQ research. Further exploration of hybrid models that combine the strengths of physical, geochemical, and ML approaches could enhance the predictive power and accuracy of GWQ models. These models should be further refined to handle complex and dynamic climate interactions, ensuring their adaptability to various hydrological conditions.

Furthermore, the analysis underscores the necessity for continuous development and refinement of these models to account for emerging climatic patterns and extreme events, which pose significant challenges to groundwater sustainability. The findings emphasize the urgency for adaptive management strategies that incorporate the predictive capabilities of advanced modelling techniques to mitigate the adverse impacts of climate change on groundwater resources.

In summary, as climate change continues to alter hydrological cycles and exacerbate environmental stressors, the integration of ML with analytical models represents a promising avenue for enhancing the resilience of groundwater systems. This approach improves the accuracy of predictions and equips stakeholders with the necessary tools to safeguard GWQ in an era of unprecedented climatic variability. Future research should focus on refining these models, emphasizing the need for high-resolution

data and cross-disciplinary collaborations to address the complex challenges posed by climate-induced changes in GWQ.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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Appendix

Table A1. Paper per country with the continental approach.

Continents	Articles
Europe	315
Austria, Portugal	12
Belgium, Turkey	8
Croatia, United Kingdom	4
Czech Republic, Finland, Poland, Wales	5
Spain, Denmark	15, 16
Egypt, Romania	6
Italy, England	27, 28
Netherlands, France	18, 22
Germany	59
Hungary, Slovenia, Scotland	3
Ireland, Greece, Latvia, Monaco, Slovakia	1
Sweden, Switzerland	10, 20
Estonia, Norway, Georgia	2
Asia	190
China	70
India, Iran	37, 15
Bangladesh, South Korea	6
Malaysia	9
Japan, Pakistan, VietNam, Russia	5
Singapore, Saudi Arabia, U Arab Emirates	4
Oman	3
Philippines, Indonesia, Kuwait, Iraq	2
Qatar, Bahrain, Sri Lanka, Thailand	1
America	162
United States, Canada	117, 29
Brazil, Argentina	5, 2
Chile, Mexico	3
Nicaragua, Colombia, Peru	1
Africa	30
Benin, Botswana, Cameroon, Ethiopia,	1
Ghana, Kenya, Tunisia, Morocco	2, 3
Uganda, Senegal	4
Zambia, Nigeria	9
South Africa	
Australia	47
Antarctica	0

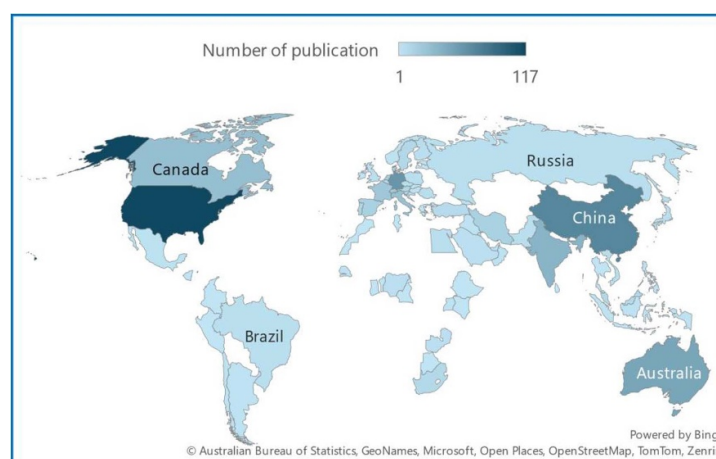


Figure A1. Publications per country in the literature search.

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